MBG 1201 Introduction to Modern Biology

Course Lecturer: Şebnem Öztürk

seb.ozturk@gmail.com

sebnemozturk@marun.edu.tr

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The Relationship Between Dominance and Phenotype

- An allele is called dominant because it is seen in the phenotype, not because it somehow subdues a recessive allele.

- Alleles are simply variations in a gene's nucleotide sequence.

- When a dominant allele coexists with a recessive allele in a heterozygote, they do not interact at all.

It is in the pathway from genotype to phenotype that dominance and recessiveness come into play.

Tay-Sachs disease, an inherited disorder in humans

For any character, the observed dominant/recessive relationship of alleles depends on the level at which we examine phenotype.

• Tay-Sachs disease is fatal; a dysfunctional enzyme causes an accumulation of lipids in the brain

– Only children who inherit two copies of the Tay-Sachs allele (homozygotes) have the disease. Thus, at the organismal level, the Tay-Sachs allele qualifies as recessive.

– At the biochemical level, the phenotype (i.e., the enzyme activity level) is incompletely dominant. The activity level of the lipid-metabolizing enzyme in heterozygotes is intermediate between the activity level in individuals homozygous for the normal allele and the activity level in individuals with Tay-Sachs disease.

– At the molecular level, the alleles are codominant. Heterozygous individuals produce equal numbers of normal and dysfunctional enzyme molecules.

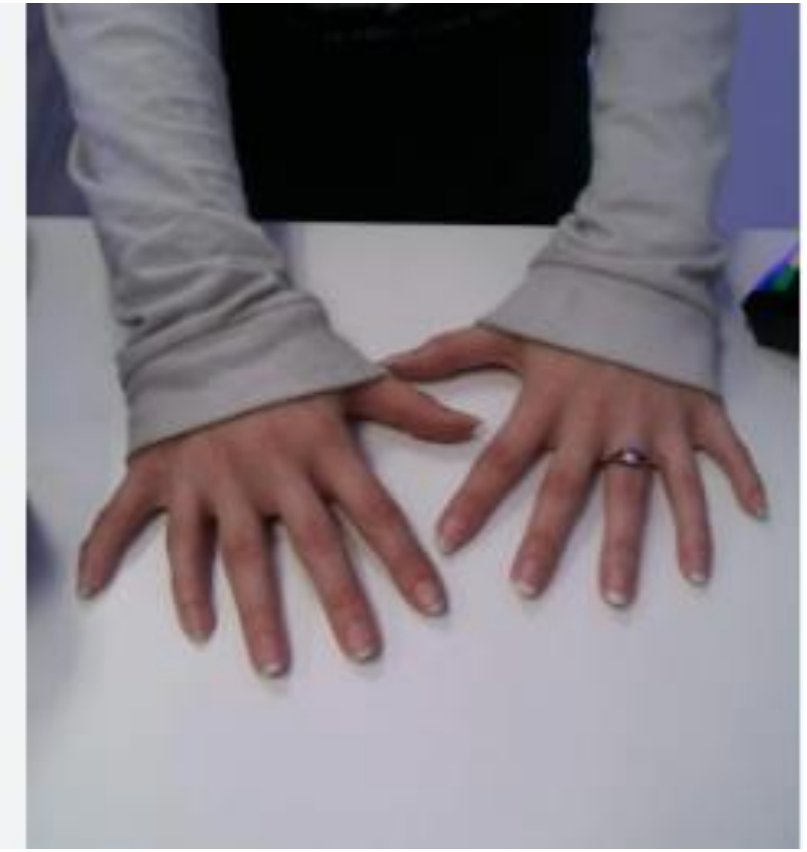
Frequency of Dominant Alleles

- Dominant alleles are not necessarily more common in populations than recessive alleles

- For example, one baby out of 400 in the United States is born with extra fingers or toes

- The allele for this unusual trait is dominant to the allele for the more common trait of five digits per appendage

- In this example, the recessive allele is far more prevalent than the population's dominant allele



Multiple Alleles

• Most genes exist in populations in more than two allelic forms

• For example, the four phenotypes of the ABO blood group in humans are determined by three alleles for the enzyme (I) that attaches A or B carbohydrates to red blood cells: IA, IB, and i.

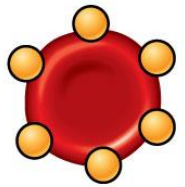
• The enzyme encoded by the IA allele adds the A carbohydrate, whereas the enzyme encoded by the IB allele adds the B carbohydrate; the enzyme encoded by the i allele adds neither

Both the IA and IB alleles are dominant to the i allele.

(a) The three alleles for the ABO blood groups and their carbohydrates

Allele	I^A	I^B	i
Carbohydrate	A	B	none

(b) Blood group genotypes and phenotypes

Genotype	I^AI^A or I^Ai	I^BI^B or I^Bi	I^AI^B	ii
Red blood cell appearance				
Phenotype (blood group)	A	B	AB	O

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Extending Mendelian Genetics for Two or More Genes

- Most genes have multiple phenotypic effects, a property called **pleiotropy**

- For example, pleiotropic alleles are responsible for the multiple symptoms of certain hereditary diseases, such as cystic fibrosis and sickle-cell disease

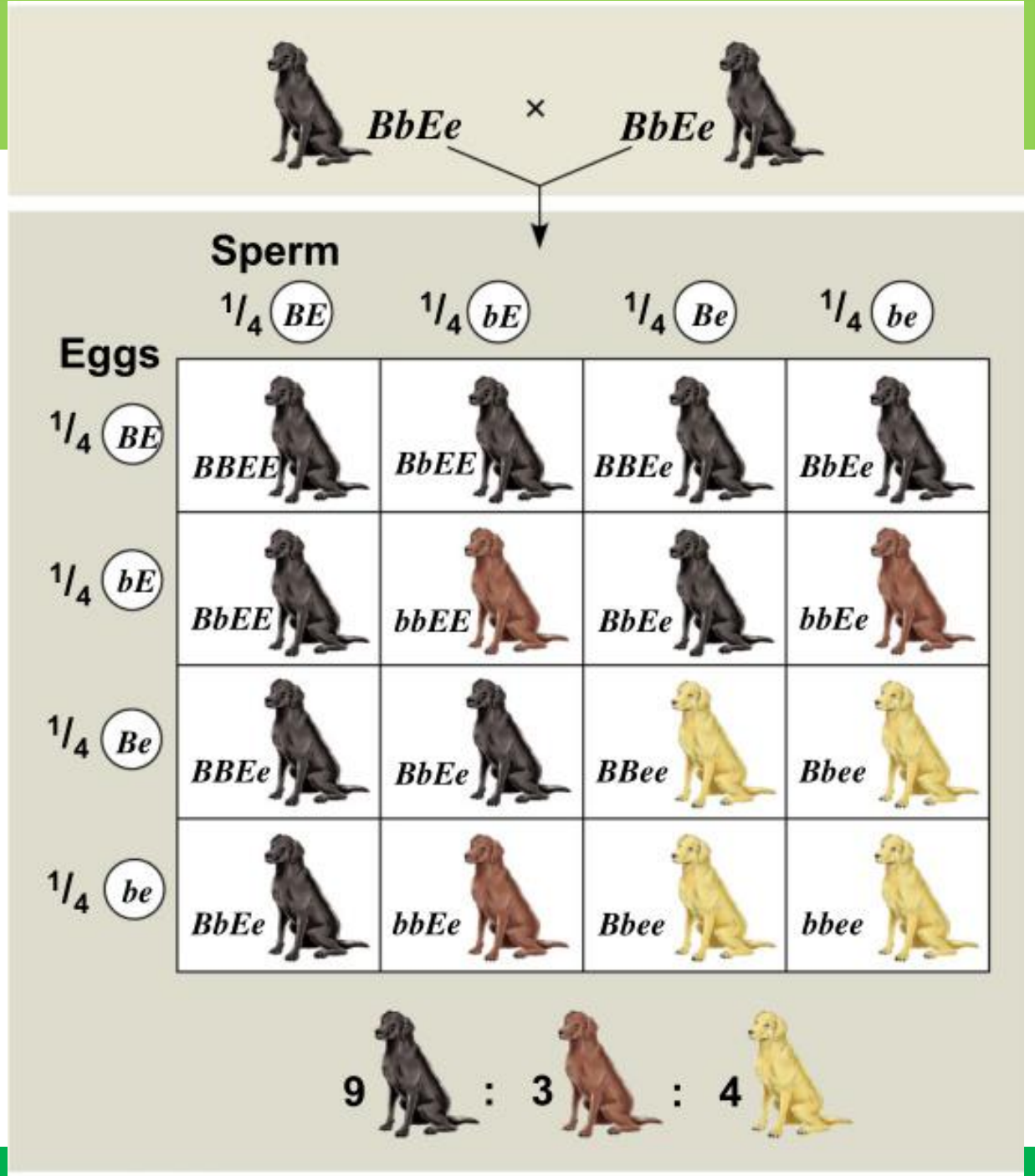
- Some traits may be determined by two or more genes

- In **epistasis**, a gene at one locus alters the phenotypic expression of a gene at a second locus.

e.g. Coat color in Labrador retrievers depends on two genes.
1 gene determines pigment color (B for black, and b for brown)
1 gene determines (C: color; c: no color) whether the pigment will be deposited in the hair

Epistasis

An example of epistasis. This Punnett square illustrates the genotypes and phenotypes predicted for offspring of matings between two black Labrador retrievers of genotype BbEe. The E/e gene, which is epistatic to the B/b gene coding for hair pigment, controls whether or not pigment of any color will be deposited in the hair.



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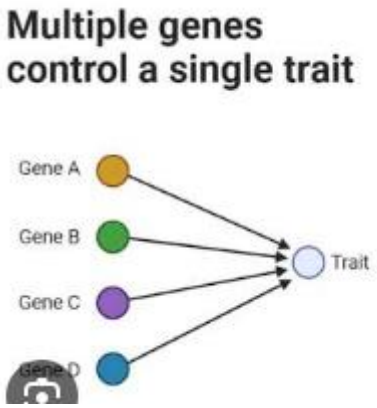
Polygenic Inheritance

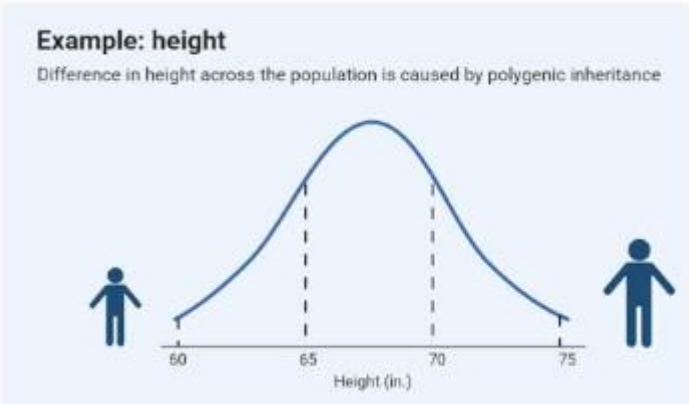
•Polygenic inheritance : quantitative inheritance, where multiple independent genes have an additive or similar effect on a single quantitative trait.

Quantitative characters are those that vary in the population in a gradual fashion
e.g. height

• Quantitative variation usually indicates **polygenic inheritance** an additive effect of two or more genes on a single phenotype

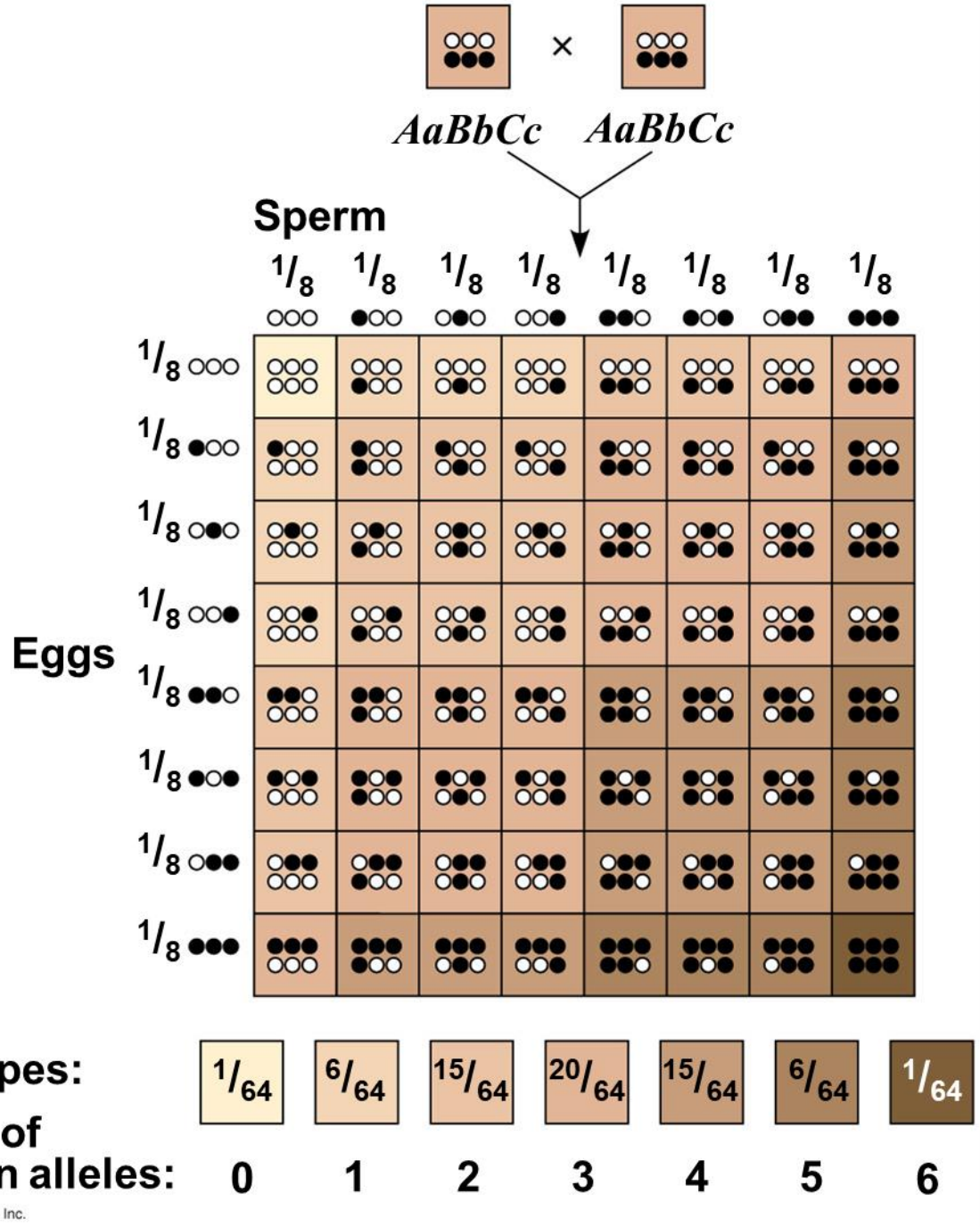
In 2014, a genomic study of over 250,000 people found almost 700 genetic variations in over 180 genes that affect height.





Polygenic Inheritance

A simplified model for polygenic inheritance of skin color. In this model, three separately inherited genes affect skin color. (The reported number is actually 378 genes.) The heterozygous individuals (AaBbCc) represented by the two rectangles at the top of this figure each carry three dark-skin alleles (black circles, which represent A, B, or C) and three light- skin alleles (white circles, which represent a, b, or c). The Punnett square shows all possible genetic combinations in gametes and offspring of hypothetical matings between these two. The results are summarized by phenotypic frequencies (fractions) under the Punnett square. (The phenotypic ratio is 1:6:15:20:15:6:1.)



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The Environmental Impact on Phenotype

• Another departure from Mendelian genetics arises when the phenotype for a character depends on environment as well as genotype.

• Traits that depend on multiple genes combined with environmental influences are called **multifactorial.** Such characters are called multifactorial because genetic and environmental factors collectively influence phenotype

• The **norm of reaction** is the phenotypic range of a genotype influenced by the environment.
Norms of reaction are generally broadest for polygenic characters
• For example, hydrangea flowers of the same genotype range from blue-violet to pink, depending on soil acidity



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Integrating a Mendelian View of Heredity and Variation

- An organism's phenotype includes its physical appearance, internal anatomy, physiology, and behavior

- An organism's phenotype reflects its overall genotype and unique environmental history

Many human traits follow Mendelian patterns of inheritance

- Humans are not good subjects for genetic research

– Generation time is too long

– Parents produce relatively few offspring

– Breeding experiments are unacceptable

- However, basic Mendelian genetics endures as the foundation of human genetics

Pedigree Analysis

• A **pedigree** is a family tree that describes the interrelationships of parents and children across generations

•Inheritance patterns of particular traits can be traced and described using pedigrees

• Information about the presence or absence of a particular phenotypic trait is collected from as many individuals in a family as possible, across generations.

• The distribution of these characters is then mapped on the family tree.

An important application of a pedigree is to help us calculate the probability that a future child will have a particular genotype and phenotype.

Pedigree Analysis

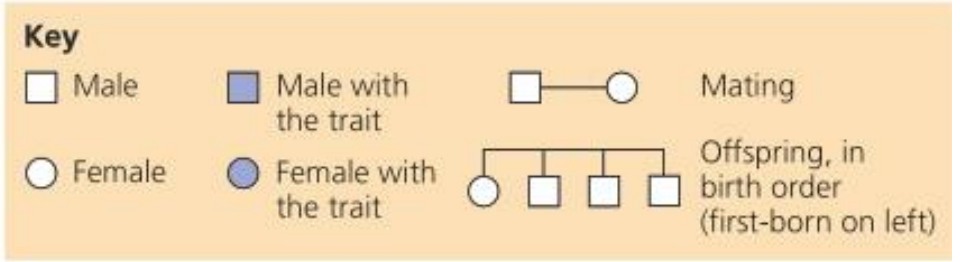
– For example, the occurrence of a widow's peak (W) is dominant to a straight hairline (w).

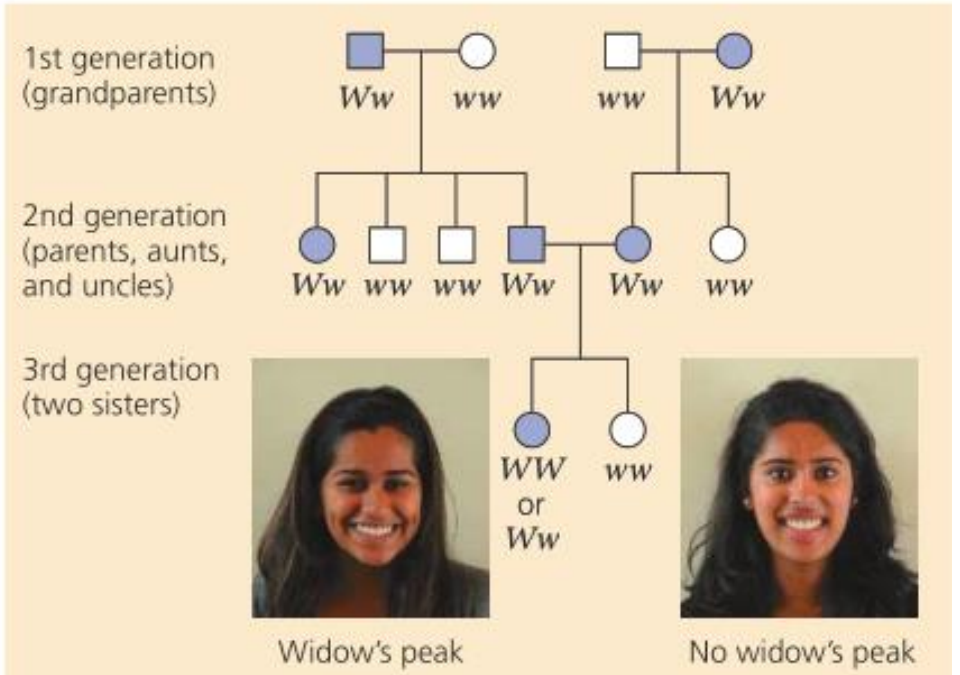
• Phenotypes of family members and knowledge of dominance/recessiveness relationships between alleles allow researchers to predict the genotypes of members of this family.

– For example, if an individual in the third generation lacks a widow's peak but both her parents have widow's peaks, then her parents must be heterozygous for that gene.

– If some siblings in the second generation lack a widow's peak and one of the grandparents (first generation) also lacks one, then we know the other grandparent must be heterozygous, and we can determine the genotype of many other individuals.

▼ Figure 14.15 Pedigree analysis. Each pedigree traces a trait through three generations of the same family; the two traits have different inheritance patterns. For the sake of understanding genetic principles, each trait is shown as being determined by two alleles of a single gene—a simplification, since other genes likely also affect these two characters.





(a) Is a widow's peak a dominant or recessive trait?
Tips for pedigree analysis: Notice in the third generation that the second-born daughter lacks a widow's peak, although both of her parents had the trait. Such a pattern indicates that the trait is due to a dominant allele. If it were due to a *recessive* allele, and both parents had the recessive phenotype (straight hairline), *all* of their offspring would also have the recessive phenotype.

Recessively Inherited Disorders

- Many genetic disorders are inherited in a recessive manner.
- These range from relatively mild to life-threatening
- Recessively inherited disorders show up only in individuals homozygous for the recessive allele
- **Carriers** are heterozygous individuals who carry the recessive allele but are phenotypically normal

- If a recessive allele that causes a disease is rare, then the chance of two carriers meeting and mating is low

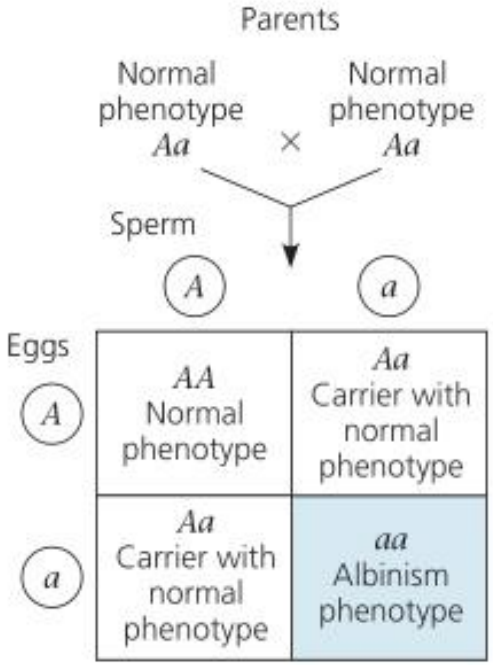
- Consanguineous matings (i.e., matings between close relatives) increase the chance of mating between two carriers of the same rare allele

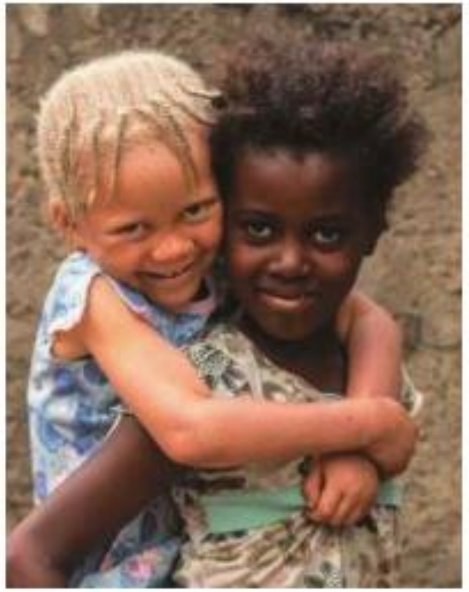
- Most societies and cultures have laws or taboos against marriages between close relatives

Recessively Inherited Disorders

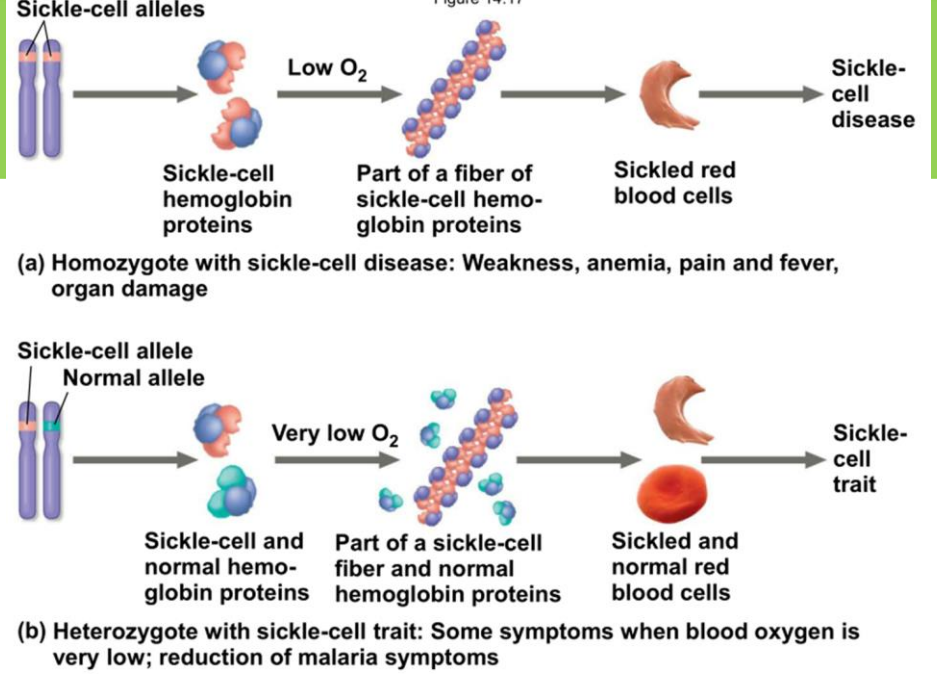
Genes code for proteins of specific function. An allele that causes a genetic disorder (let's call it allele a) codes for either a malfunctioning protein or no protein at all. In recessive disorders, heterozygotes (Aa) typically have the normal phenotype because one copy of the normal allele (A) produces a sufficient amount of the specific protein. Thus, a recessively inherited disorder shows up only in the homozygous individuals (aa) who inherit a recessive allele from each parent. Although phenotypically normal with regard to the disorder, heterozygotes may transmit the recessive allele to their offspring and thus are called carriers.

- Albinism is a recessive condition characterized by a lack of pigmentation in skin and hair
- Most recessive homozygotes are born to parents who are carriers of the disorder, but themselves lack the disorder





Sickle-Cell Disease



- The disease is caused by the substitution of a single amino acid in the hemoglobin protein in red blood cells. Glutamine is replaced by Valine.
- The non sickle allele is incompletely dominant to the sickle-cell allele.
- Heterozygotes (said to have sickle-cell trait) are usually healthy but may suffer symptoms of sickle-cell disease under blood-oxygen stress.
- About one out of ten African Americans has sickle cell trait, an unusually high frequency of an allele with detrimental effects in homozygotes
- Heterozygotes are less susceptible to the malaria parasite (Heterozygous advantage).

Even a slight heterozygote advantage may act to increase the frequency of the mutant allele in the population—even if the mutant allele causes major reduction in fitness in homozygotes in that population.

Homozygotes (AA) for the normal allele are susceptible to malaria; many become infected and some die, leading to reduced fitness. SS homozygotes are even more reproductively disadvantaged because of their severe hematologic disease, their fitness approaching zero. Heterozygotes for sickle cell disease have red blood cells that resist infection by the malaria parasite and burst more quickly when infected, thereby leading to the death of the parasites before they infect other cells.

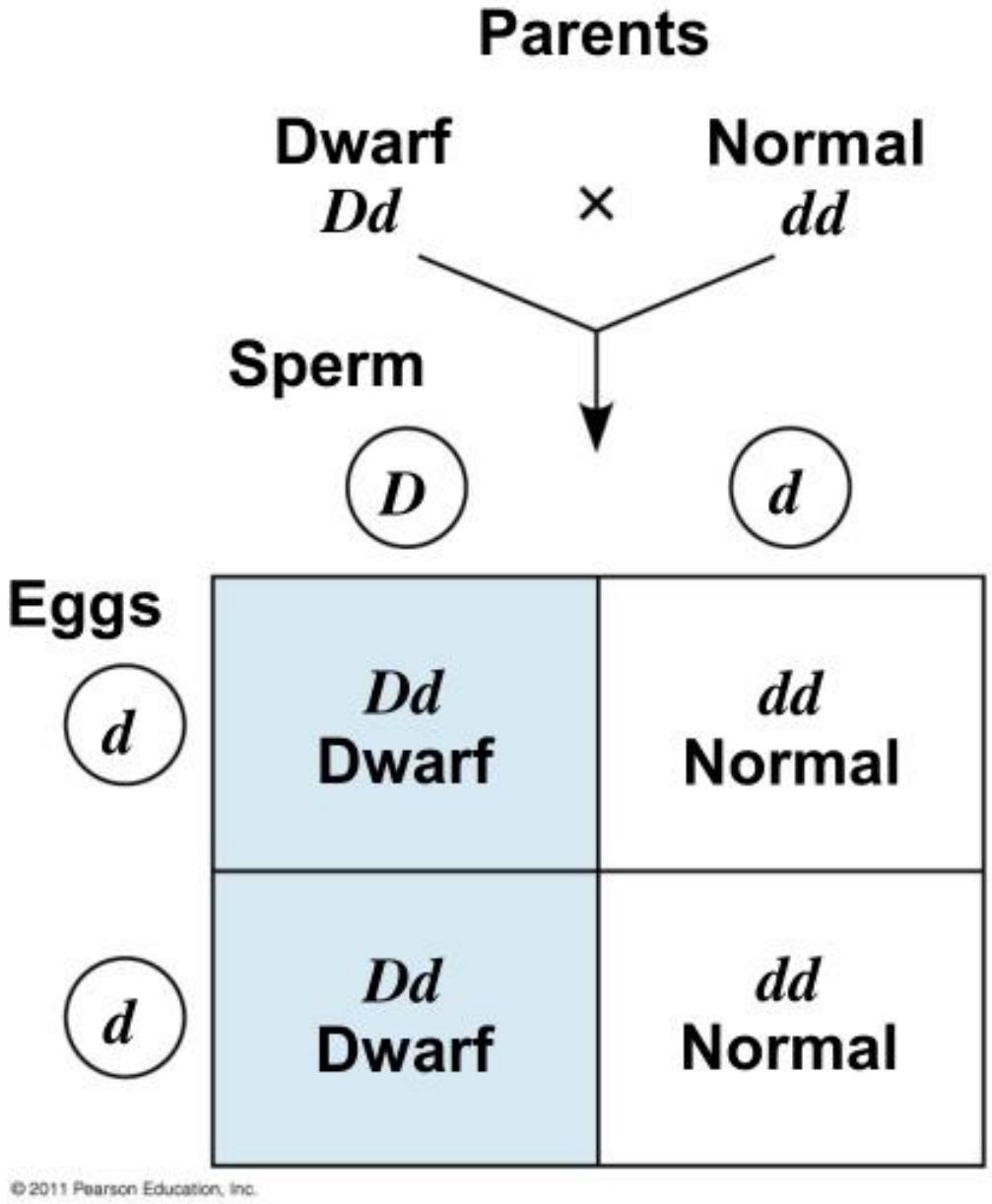
Dominantly Inherited Disorders

- Some human disorders are caused by dominant alleles
- Dominant alleles that cause a lethal disease are rare and arise by mutation
- Achondroplasia is a form of dwarfism caused by a rare dominant allele

– Heterozygous individuals have the dwarf phenotype.

– The 99.99% of the population who are not achondroplastic dwarfs are homozygous recessive for this trait.

– Achondroplasia is another example of a trait for which the recessive allele is far more prevalent than the dominant allele.



Dominantly Inherited Disorders

- A lethal dominant allele can escape elimination if it causes death at a relatively advanced age, after the individual has already passed on the lethal allele to his or her children.
- One example is Huntington's disease, a degenerative disease of the nervous system (1:10,000 people)
  - The dominant lethal allele has no obvious phenotypic effect until the individual is about 35 to 45 years old.
  - Then the deterioration of the nervous system is irreversible and inevitably fatal.
- Any child born to a parent who has the allele for Huntington's disease has a 50% chance of inheriting the disease and the disorder.
- Recently, molecular geneticists have used pedigree analysis of affected families to track the Huntington's allele to a locus near the tip of chromosome 4.
- The gene has now been sequenced.
- This has led to the development of a test that can detect the presence of the Huntington's allele in an individual's genome.

Multifactorial Disorders

- Many diseases, such as heart disease, diabetes, alcoholism, mental illnesses, and cancer have both genetic and environmental components
- No matter what our genotype, our lifestyle has a tremendous effect on phenotype
- Little is understood about the genetic contribution to most multifactorial diseases

Counseling Based on Mendelian Genetics and Probability Rules

- Genetic counselors can provide information to prospective parents concerned about a family history for a specific disease
- Using family histories, genetic counselors help couples determine the odds that their children will have genetic disorders
- It is important to remember that each child represents an independent event in the sense that its genotype is unaffected by the genotypes of older siblings
- For a growing number of diseases, tests are available that identify carriers and help define the odds more accurately
- The tests enable people to make more informed decisions about having children
- However, they raise other issues, such as whether affected individuals fully understand their genetic test results
- Probabilities are predicted on the most accurate information at the time; predicted probabilities may change as new information is available

In Utero Fetal Testing

- In amniocentesis, the liquid that bathes the fetus is removed and tested usually between 14-16 weeks
  – Amniocentesis can be used to assess whether the fetus has a specific disease.
- Fetal cells extracted from amniotic fluid are cultured and karyotyped to identify some disorders.
- Other disorders can be identified from chemicals in the amniotic fluids.
- In chorionic villus sampling (CVS), a sample of the placenta is removed and tested
- Other techniques, such as ultrasound and fetoscopy, allow fetal health to be assessed visually in utero

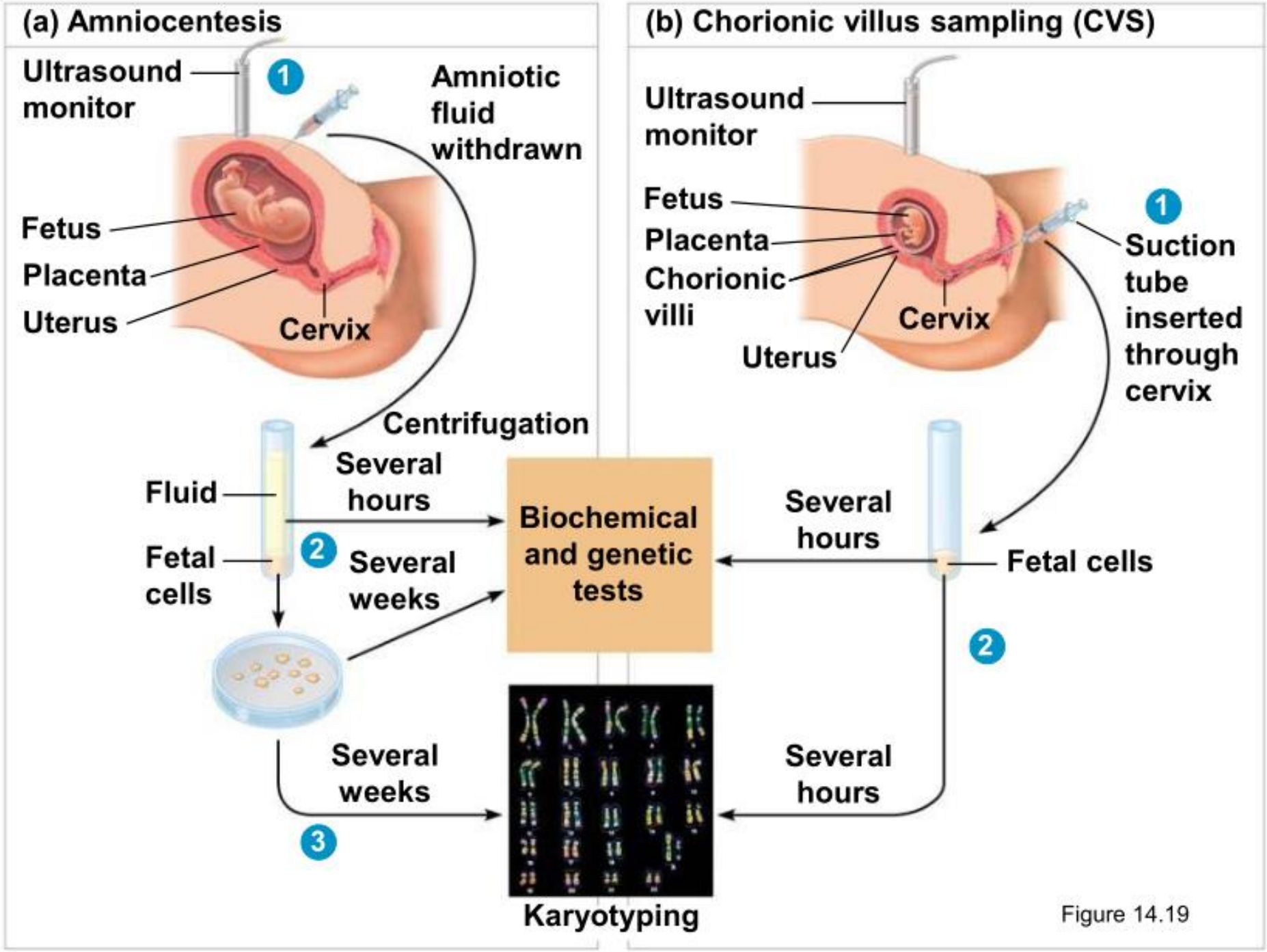

(a) Amniocentesis
Ultrasound monitor
1
Amniotic fluid withdrawn
Fetus
Placenta
Uterus
Cervix
Centrifugation
Fluid
Several hours
Fetal cells
2
Several weeks
Biochemical and genetic tests
Several weeks
3
Karyotyping
(b) Chorionic villus sampling (CVS)
Ultrasound monitor
Fetus
Placenta
Chorionic villi
Uterus
Cervix
1
Suction tube inserted through cervix
Several hours
Fetal cells
2
Several hours

Figure 14.19

Newborn Screening

- Some genetic disorders can be detected at birth by simple tests

- One common test is for phenylketonuria (PKU), a recessively inherited disorder that occurs in one of every 10,000–15,000 births in the United States



Relationship among alleles of a single gene	Description	Example
Complete dominance of one allele	Heterozygous phenotype same as that of homozygous dominant	
Incomplete dominance of either allele	Heterozygous phenotype intermediate between the two homozygous phenotypes	
Codominance	Both phenotypes expressed in heterozygotes	
Multiple alleles	In the whole population, some genes have more than two alleles	ABO blood group alleles I^A, I^B, i
Pleiotropy	One gene is able to affect multiple phenotypic characters	Sickle-cell disease

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Chromosome Theory of Inheritance (~1902)

- The chromosome theory of inheritance states:

–Mendelian genes have specific loci (positions) on chromosomes

–Chromosomes undergo segregation and independent assortment

- The behavior of chromosomes during meiosis was said to account for Mendel's laws of segregation and independent assortment

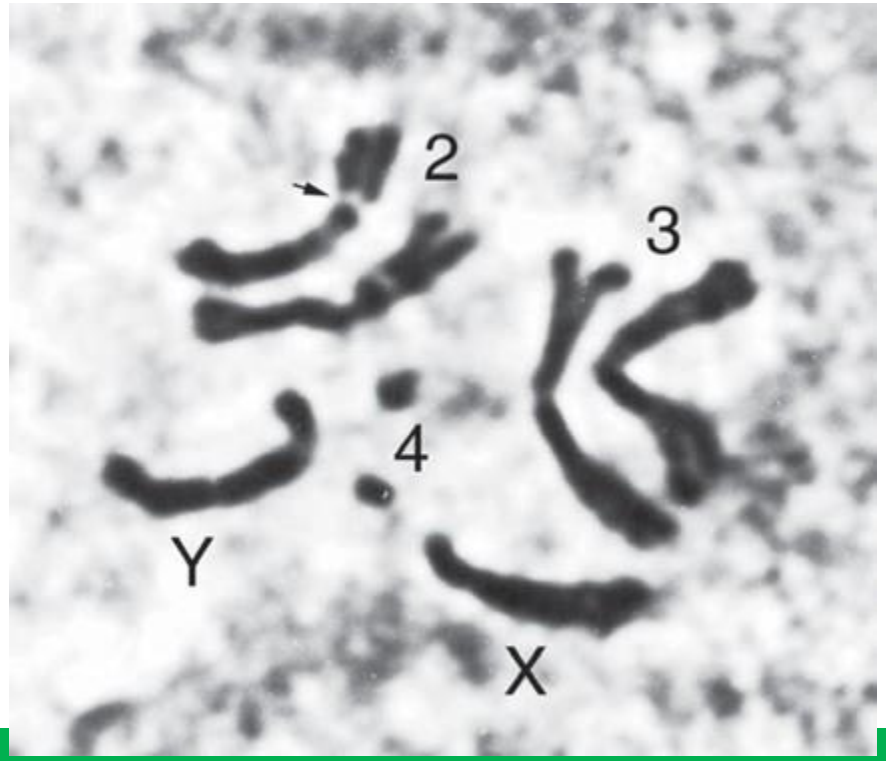
Morgan's Experimental Evidence

• The first solid evidence associating a specific gene with a specific chromosome came from Thomas Hunt Morgan, an experimental embryologist at Columbia University.

• Morgan's experiments with Drosophila melanogaster (fruit flies) provided convincing evidence that chromosomes are the location of Mendel's heritable factors.

- Several characteristics make fruit flies a convenient
- organism for genetic studies:

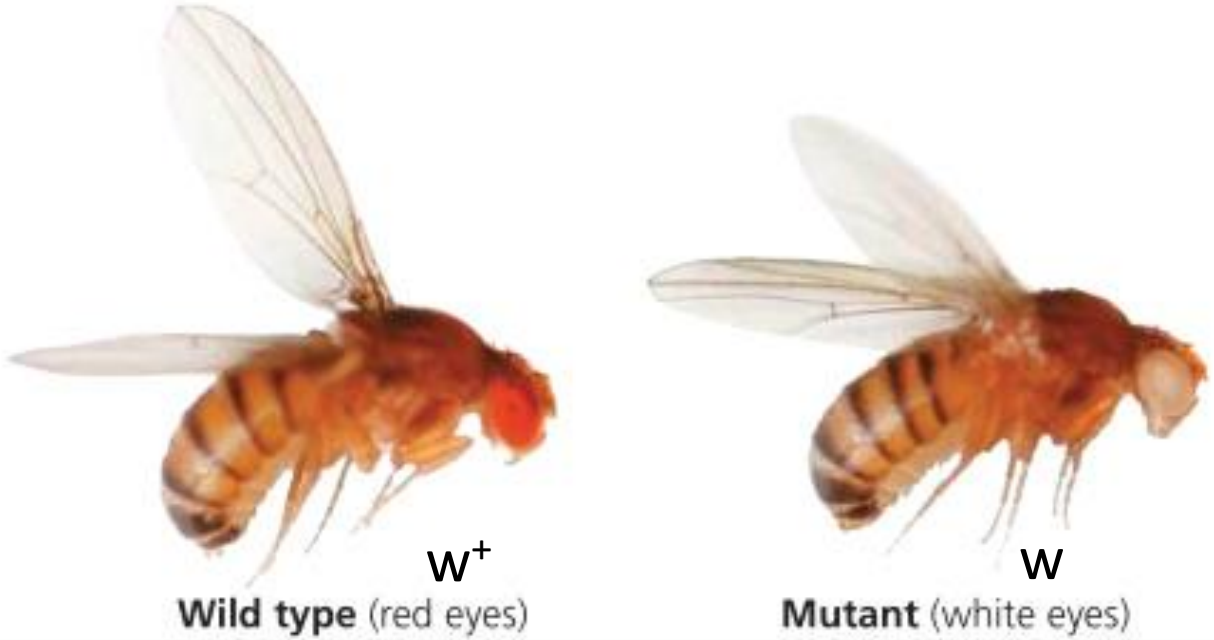
– a single mating will produce hundreds of offspring
– a new generation can be bred every two weeks
–have only four pairs of chromosomes which are easily distinguishable with a light microscope.



Morgan's Experimental Evidence

Morgan noted **wild type, or normal, phenotypes** that were common in the fly populations
Traits alternative to the wild type are called **mutant phenotypes** (because they are due to alleles assumed to have originated as changes, or mutations, in the wild-type allele).

– Morgan spent a year looking for variant individuals among the flies he was breeding!
– Finally he discovered a single male fly with white eyes instead of the usual red.



For a given character in flies, the gene takes its symbol from the first mutant (non–wild type) discovered. Thus, the allele for white eyes in Drosophila is symbolized by w. A super-script + identifies the allele for the wild-type trait: w^+ for the allele for red eyes, for example. Over the years, a variety of gene notation systems have been developed for different organisms.

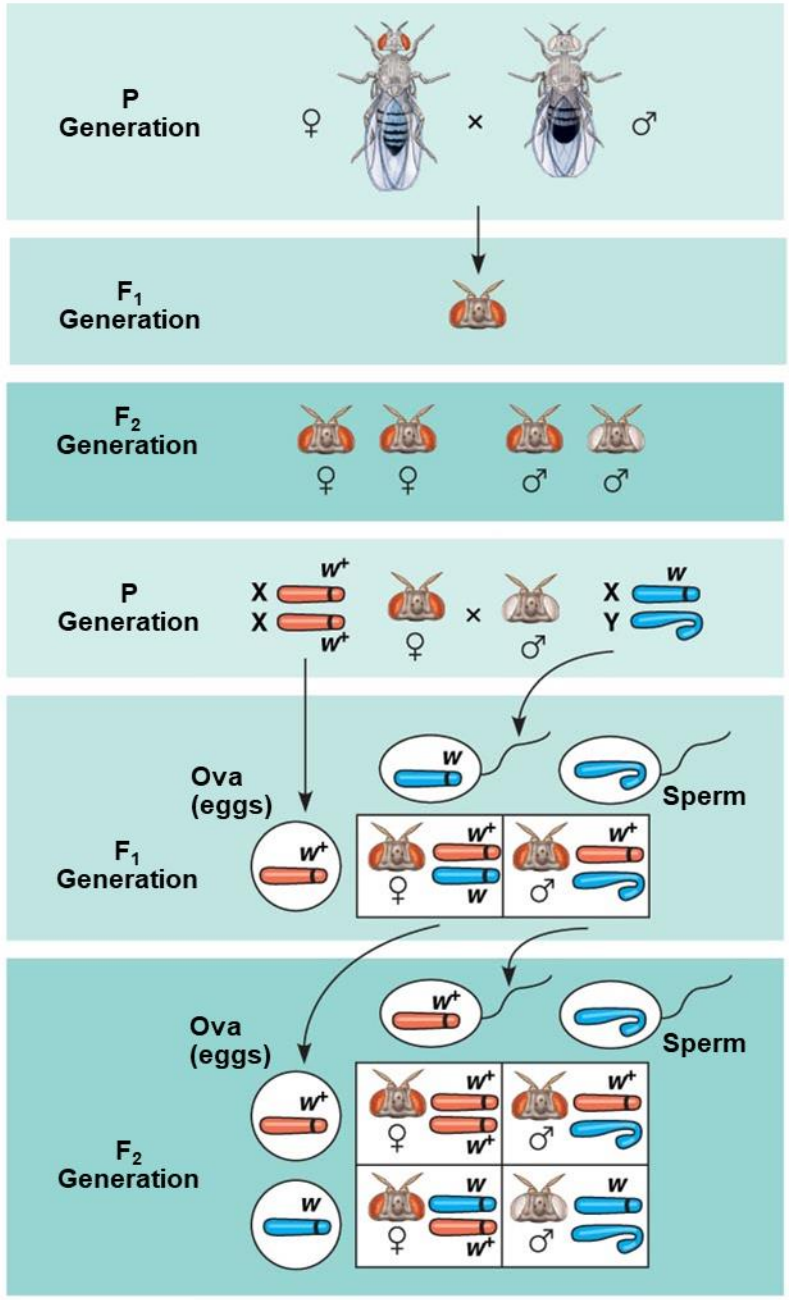
Correlating Behavior of a Gene's Alleles with Behavior of a Chromosome Pair

- Morgan mated his white-eyed male fly with a red-eyed female.
- F1 offspring had red eyes, suggesting that the wild-type allele is dominant.
- The F2 generation showed the 3:1 red:white eye ratio, but only males had white eyes

Conclusion: Somehow a fly's eye color was linked to its sex. The gene with the white-eyed mutation is on the X chromosome, with no corresponding allele present on the Y chromosome.

Females homozygous (XX) or heterozygous (Xx) have red eyes.

Males with (XY) have red eyes but (xY) have white eyes. Single X chromosome is enough to determine eye color for male flies.



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The relationship between the chromosome theory of inheritance and Mendel's laws: Chromosomes during F1 Meiosis

The two alleles for each gene separate during gamete formation.

R and r alleles segregate at anaphase I, yielding two types of daughter cells.

Each gamete gets either the R or r allele.

During fertilization, the R and r alleles recombine randomly.

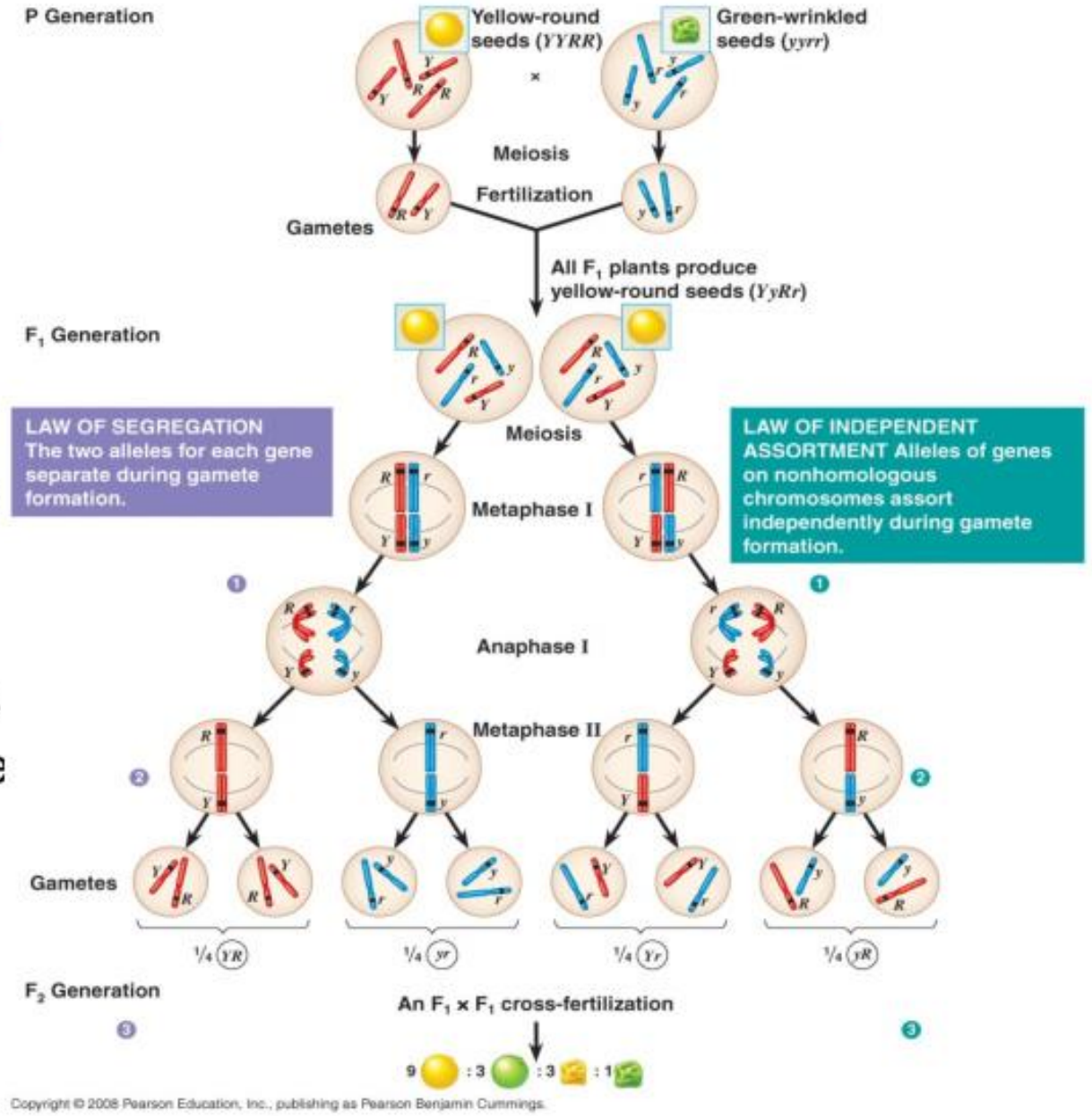


Figure15.2

Alleles on nonhomologous chromosomes assort independently during gamete formation.

1.Alleles at both loci segregate during anaphase I giving rise to 4 different daughter cells, depending on how they arranged during metaphase I.

2. Each gamete gets a chromosome in one of 4 allele combinations.

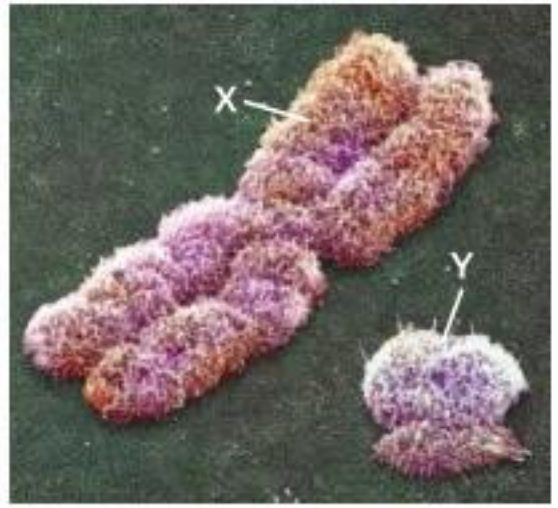
3.Fertilization results in a F2 9:3:3:1 phenotypic ratio.

Sex-linked genes exhibit unique patterns of inheritance

• In humans and some other animals, there is a chromosomal basis of sex determination. Sex in many species is determined largely by inheritance of sex chromosomes.

• Humans and other mammals have 2 types of sex chromosomes, X and Y.
– An individual who inherits two X chromosomes usually develops as a female.
– An individual who inherits an X and a Y chromosome usually develops as a male.
• Short segments at either end of the Y chromosome are the only regions that are homologous with the corresponding regions of the X. These homologous regions allow the X and Y chromosomes in males to pair and behave like homologs during meiosis.
• The SRY gene on the Y chromosome codes for a protein that directs the development of male anatomical features.
– Gonads develop into ovaries in individuals lacking SRY.

▼ **Figure 15.5** Human sex chromosomes (duplicated).



Each conception has about a fifty-fifty chance of producing a particular sex

• In both testes (XY) and ovaries (XX), the two sex chromosomes segregate during meiosis, and each gamete receives one.

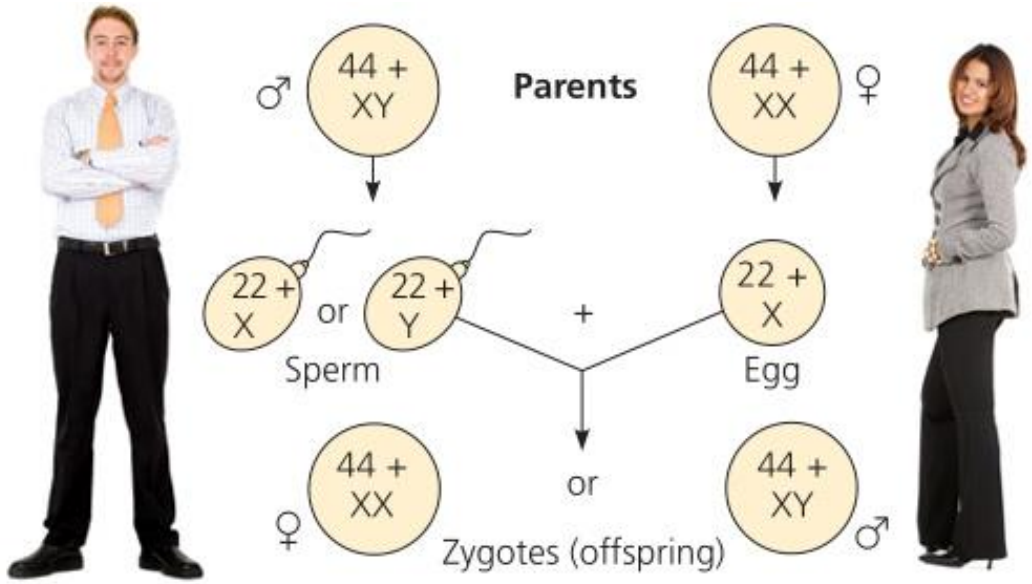
• Each ovum receives an X chromosome.

• Half the sperm cells receive an X chromosome, and half receive a Y chromosome.

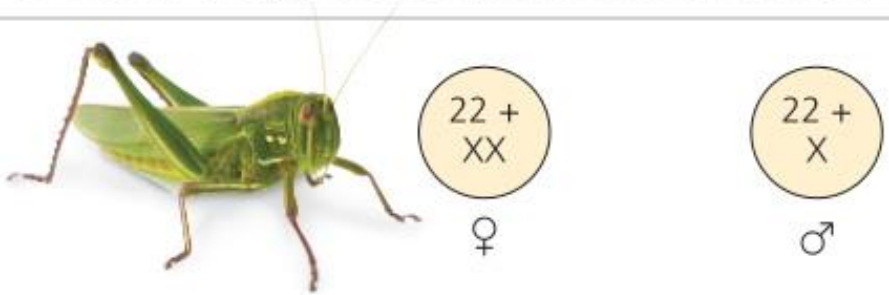
– If a sperm cell bearing an X chromosome fertilizes an ovum, the resulting zygote is female (XX).
– If a sperm cell bearing a Y chromosome fertilizes an ovum, the resulting zygote is male (XY).

--the mammalian X-Y system isn’t the only chromosomal system for determining sex.

Some chromosomal systems of sex determination



(a) The X-Y system. In mammals, the sex of an offspring depends on whether the sperm cell contains an X chromosome or a Y.



(b) The X-0 system. In grasshoppers, cockroaches, and some other insects, there is only one type of sex chromosome, the X. Females are XX; males have only one sex chromosome (X0). Sex of the offspring is determined by whether the sperm cell contains an X chromosome or no sex chromosome.



(c) The Z-W system. In birds, some fishes, and some insects, the sex chromosomes present in the egg (not the sperm) differ, and thus determine the sex of offspring. The sex chromosomes are designated Z and W. Females are ZW and males are ZZ.



(d) The haplo-diploid system. There are no sex chromosomes in most species of bees and ants. Females develop from fertilized eggs and are thus diploid. Males develop from unfertilized eggs and are haploid; they have no fathers.

Inheritance of Sex-Linked Genes

- The sex chromosomes have genes for many characters unrelated to sex.

- A gene located on either sex chromosome is called a sex-linked gene

- The human X chromosome contains approximately 1,100 genes, which are called **X-linked Genes.**

- Genes on the Y chromosome are called **Y-linked genes.** ~78 genes that code for about 25 proteins. About half of these genes are expressed only in the testis, and some are required for normal testicular functioning and the production of normal sperm.

- X chromosome have genes for many characters unrelated to sex, whereas the Y chromosome mainly encodes genes related to sex determination.

- X-linked genes in humans follow the same pattern of inheritance that Morgan observed for the eye color locus he studied in Drosophila

Inheritance of x-Linked Genes

- x-linked genes follow specific patterns of inheritance

– Fathers pass x-linked alleles to all their daughters but none of their sons

– Mothers pass x-linked alleles to both sons and daughters

- For a recessive x-linked trait to be expressed

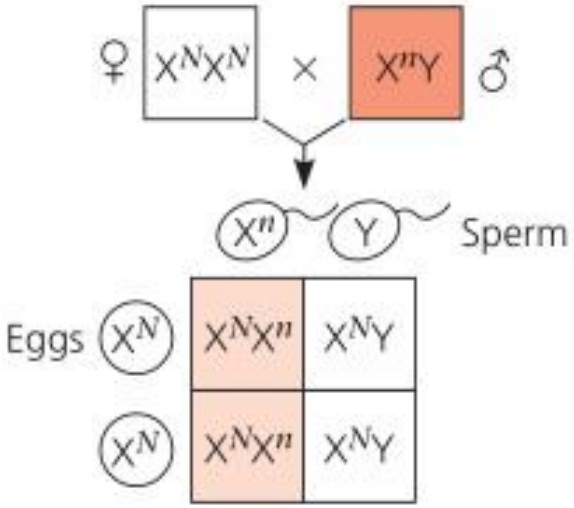
–A female needs two copies of the allele
Heterozygous females are carriers for the recessive trait.

–A male needs only one copy of the allele

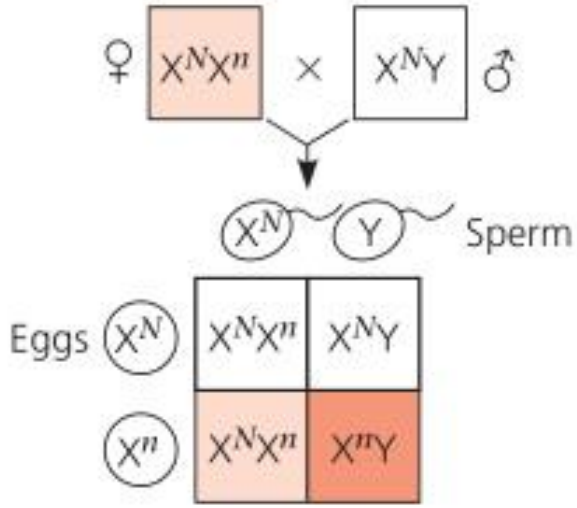
- x-linked recessive disorders are much more common in males than in females

Inheritance of Sex-Linked Genes

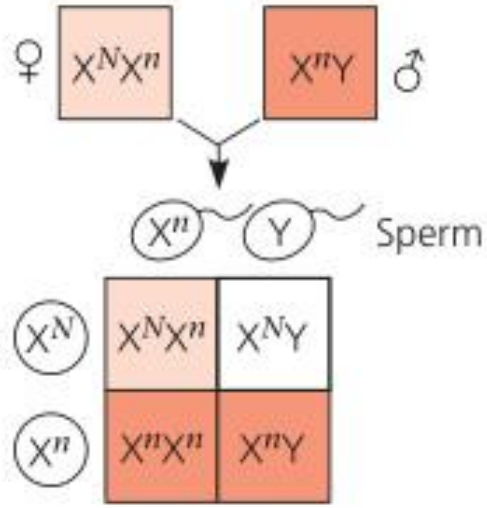
▼ Figure 15.7 The transmission of X-linked recessive traits. In this diagram, red-green color blindness is used as an example. The superscript *N* represents the dominant allele for normal color vision carried on the X chromosome, while *n* represents the recessive allele, which has a mutation causing color blindness. White boxes indicate unaffected individuals, light orange boxes indicate carriers, and dark orange boxes indicate color-blind individuals.



(a) A father with color-blindness will transmit the mutant allele to all daughters but to no sons. When the mother is a dominant homozygote, the daughters will have the normal phenotype but will be carriers of the mutation.



(b) If a carrier mates with a male who has normal color vision, there is a 50% chance that each daughter will be a carrier like her mother and a 50% chance that each son will have the disorder.



(c) If a carrier mates with a color-blind male, there is a 50% chance that each child born to them will have the disorder, regardless of sex. Daughters who have normal color vision will be carriers, whereas males who have normal color vision will be free of the recessive allele.

Inheritance of Sex-Linked Genes

- Some disorders caused by recessive alleles on the X chromosome in humans:

– **Color blindness** (mostly X-linked)

– **Duchenne muscular dystrophy**: a lethal muscular disorder
Affected individuals rarely live past their early 20s. This disorder is due to the absence of an X-linked gene for a key muscle protein called dystrophin. The disease is characterized by a progressive weakening of the muscles and a loss of coordination

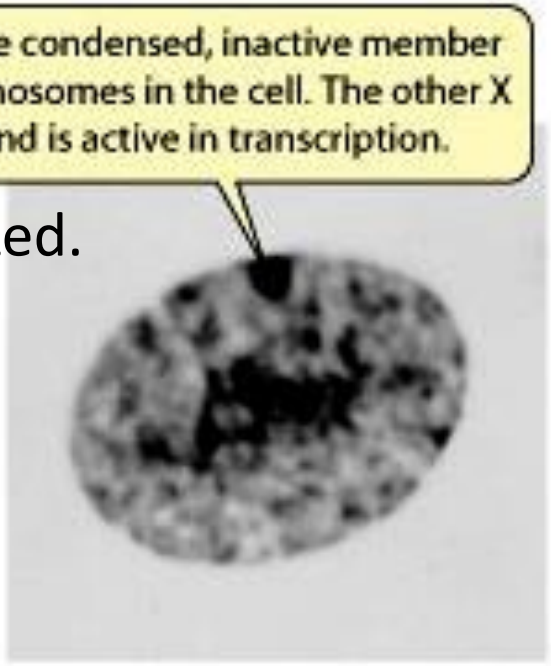
– Hemophilia: absence of one or more proteins required for blood clotting.
These proteins normally slow and then stop bleeding. Individuals with hemophilia have prolonged bleeding because a firm clot forms slowly. Bleeding in muscles and joints can be painful and can lead to serious damage. Today, people with hemophilia can be treated with intravenous injections of the missing protein.

X Inactivation in Female Mammals

- Female mammals inherit two X chromosomes—twice the number inherited by males. But females don't make twice as many of the proteins encoded by X-linked genes as males.

- One of the two X chromosomes in each cell is randomly inactivated during embryonic development. As a result, the cells of females and males have the same effective dose (one active copy) of most X-linked genes.

- The inactive X condenses into a compact object called a Barr body. During interphase an inactive X-chromosome can be visualized.



- Most of the genes on the Barr-body chromosome are not expressed.
- In the ovaries, however, Barr body chromosomes are reactivated in the cells that give rise to eggs, resulting in every egg having an active X chromosome after meiosis.

X Inactivation in Female Mammals

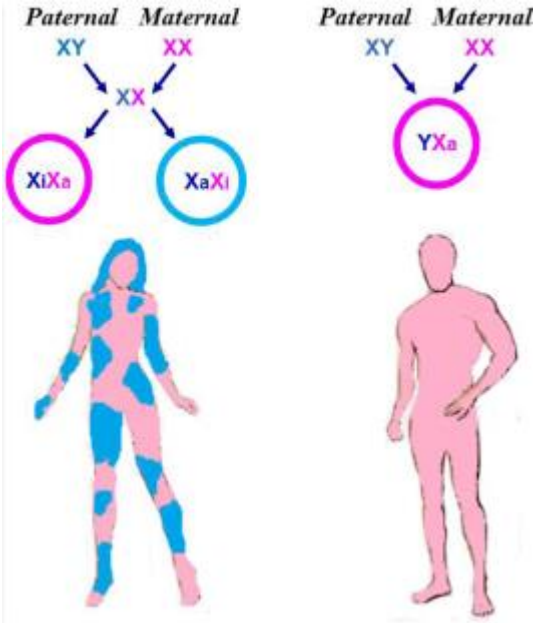
Selection of which X chromosome will form the Barr body occurs randomly and independently in embryonic cells at the time of X inactivation.

As a consequence, females consist of a mosaic of two types of cells: those with the active X derived from the father and those with the active X derived from the mother.

After an X chromosome is inactivated in a particular cell, all mitotic descendants of that cell have the same inactive X.

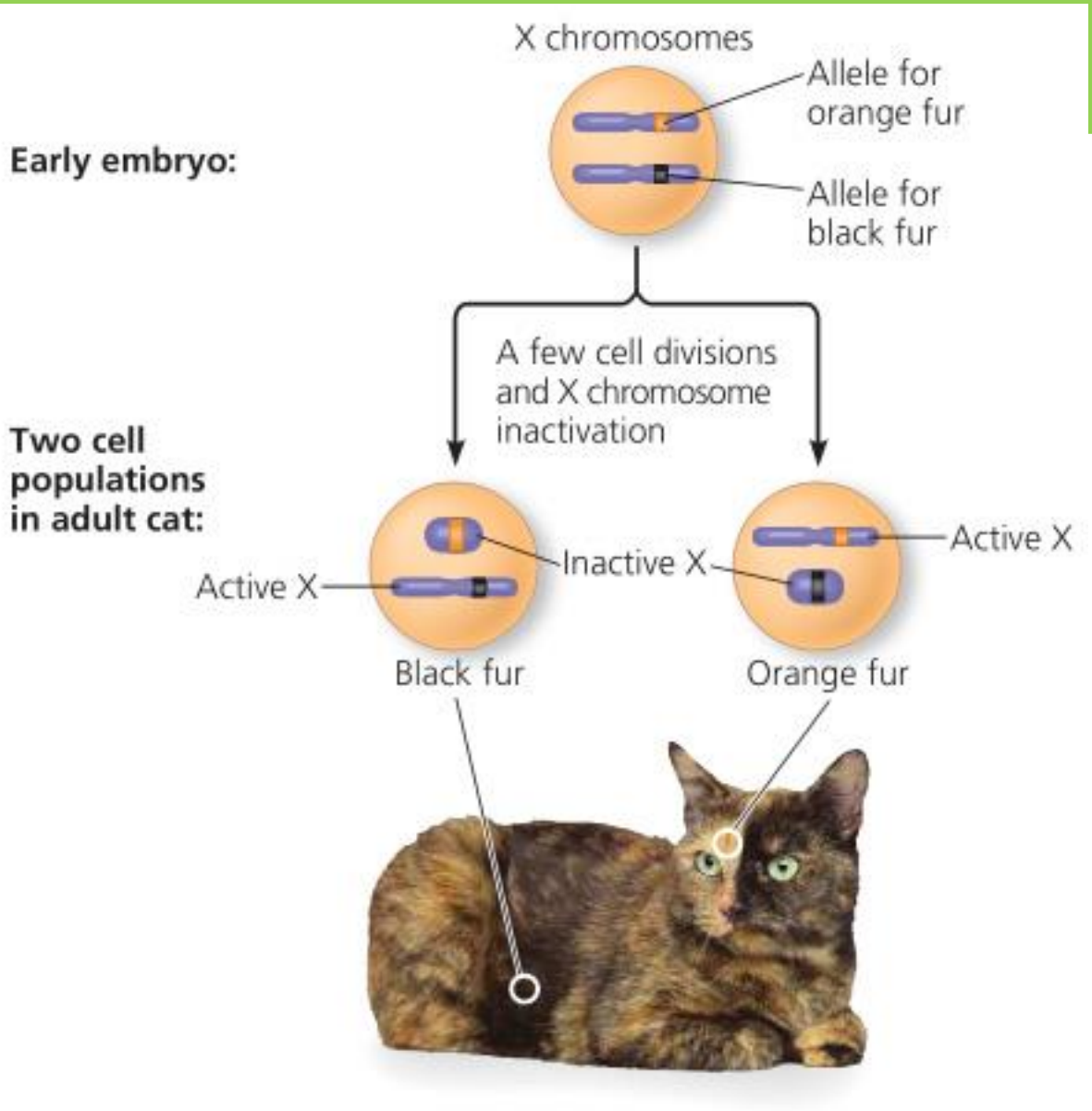
If a female is heterozygous for a sex-linked trait, approximately half her cells will express one allele, and the other half will express the other allele.

In humans, mosaicism can be observed in a recessive X-linked mutation that prevents the development of sweat glands. A woman who is heterozygous for this trait has patches of normal skin and patches of skin lacking sweat glands.



X Inactivation in the tortoiseshell cat

The tortoise shell gene is located on the X chromosome.
The tortoise-shell phenotype requires both the orange fur and black fur alleles.
Only females can have both alleles (XX).
Females heterozygous for the tortoiseshell gene have orange patches of fur where the orange allele is active as well as patches of black fur where the black allele is active.



What causes X chromosome inactivation?

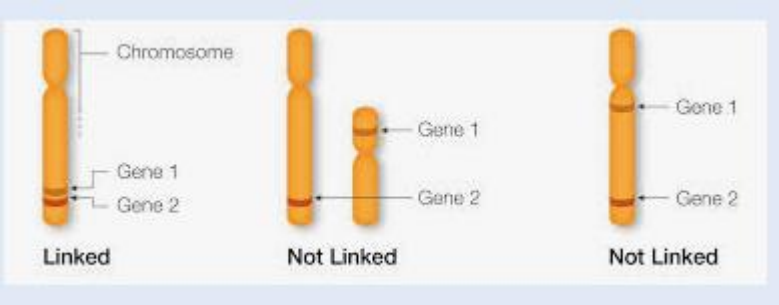
1. Methylation of DNA nucleotides of the X chromosome.

2. XIST (X-inactive specific transcript) gene.

–XIST gene is active only on the Barr-body chromosome and produces multiple copies of an RNA molecule that attach to the X chromosome on which they were made.
–This initiates X inactivation.

Linked genes tend to be inherited together

Linked genes tend to be inherited together because they are located near each other on the same chromosome.



- Each chromosome has hundreds or thousands of genes.

- Genes located near each other on the same chromosome tend to be inherited together in genetic crosses. Such genes are said to be **genetically linked** and are called **linked genes**.

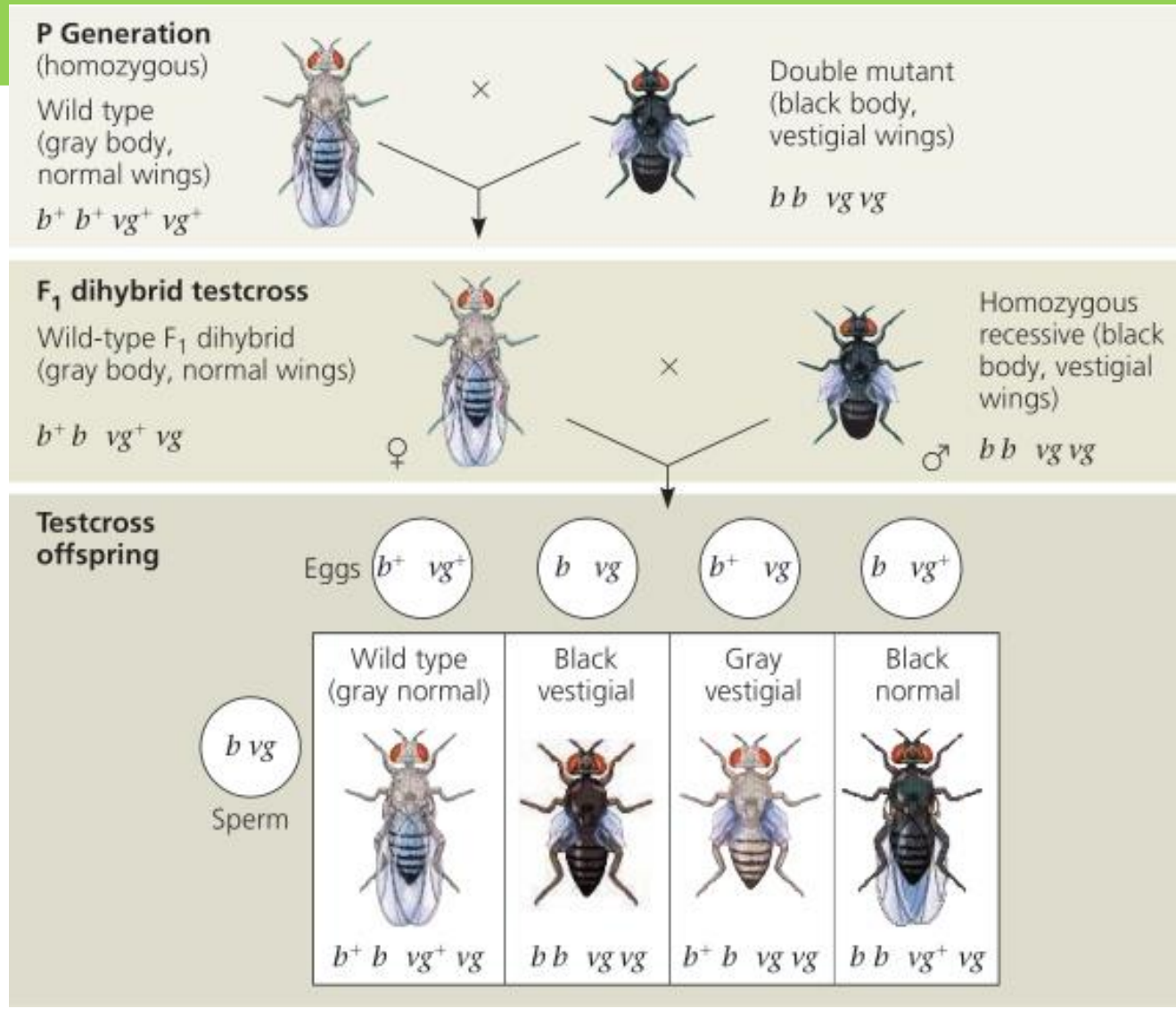
- The results of crosses with linked genes differ from the results expected according to the Mendel's law of independent assortment.

- Morgan observed this linkage and its deviations when he followed the inheritance of characters for body color and wing size in Drosophila.

How does linkage between two genes affect inheritance of characters?

Morgan wanted to know whether the genes for body color and wing size are genetically linked and, if so, how this affects their inheritance.

- Body color: wild type is gray (b+), and the mutant is black (b).
- Wing size: Wild type is (vg+), and the mutant is vestigial (vg).
  – The mutant alleles are recessive to the wild-type alleles.
  – Neither gene is on a sex chromosome.
- Morgan crossed F1 heterozygous females (b^+bvg^+vg) with homozygous recessive males (bbvgvg).
- According to independent assortment, this should produce four phenotypes in a 1:1:1:1 ratio.
- Morgan observed that most F1 offspring resembled parents



if the genes were on seperate chromosomes	1:	1:	1:	1
if genes are located on the same chromosome	1:	1:	0:	0:
Morgan's data	965:	944:	206:	185

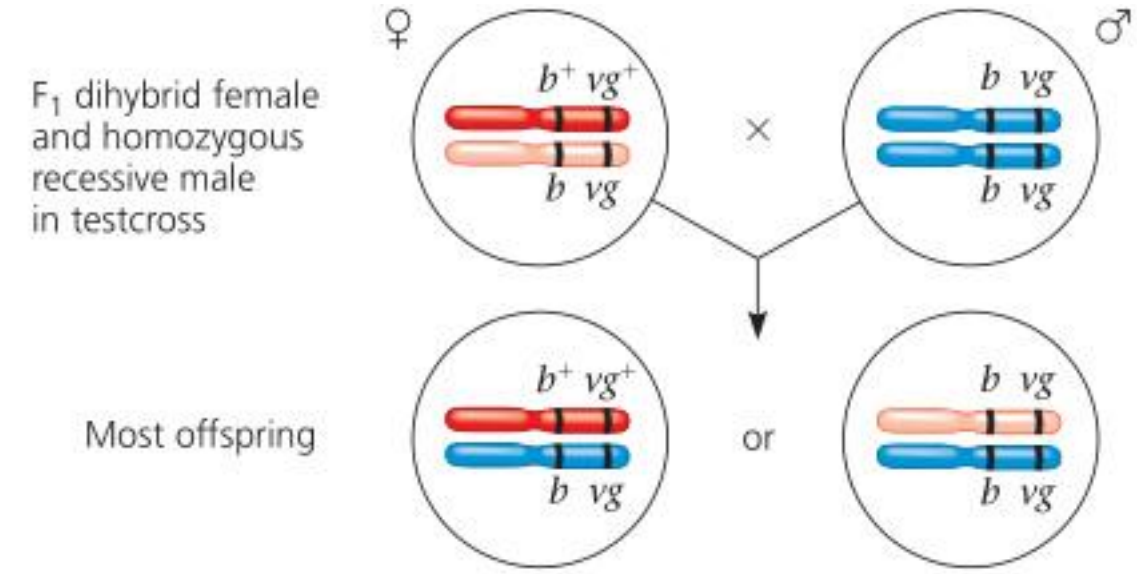
How does linkage between two genes affect inheritance of characters?

How Linkage Affects Inheritance

• Morgan found that body color and wing size are <u>usually</u> inherited together in specific combinations (parental phenotypes)

• He noted that these genes do not assort independently, and reasoned that they were on the same chromosome

• However, nonparental phenotypes were also produced. Understanding this result involves exploring genetic **recombination**, the production of offspring with combinations of traits differing from either parent.



Genetic Recombination

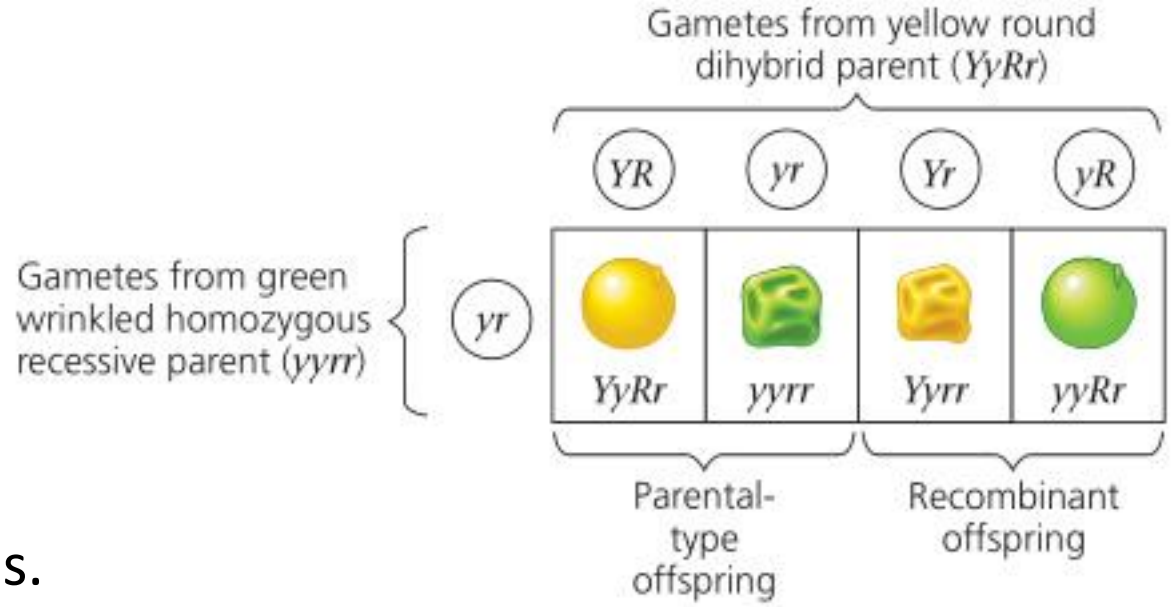
• **Genetic recombination:** The production of offspring with combinations of traits that differ from those found in either parent.
– Genetic recombination can result from independent assortment of genes located on nonhomologous chromosomes.

• Offspring with a phenotype matching one of the parental phenotypes are called **parental types**
• Offspring with nonparental phenotypes (new combinations of traits) are called **recombinant types**, or recombinants

• A 50% frequency of recombination (50% of all offspring are recombinants) is observed for any two genes on different chromosomes.

Recombination of Unlinked Genes: Independent Assortment of Chromosomes

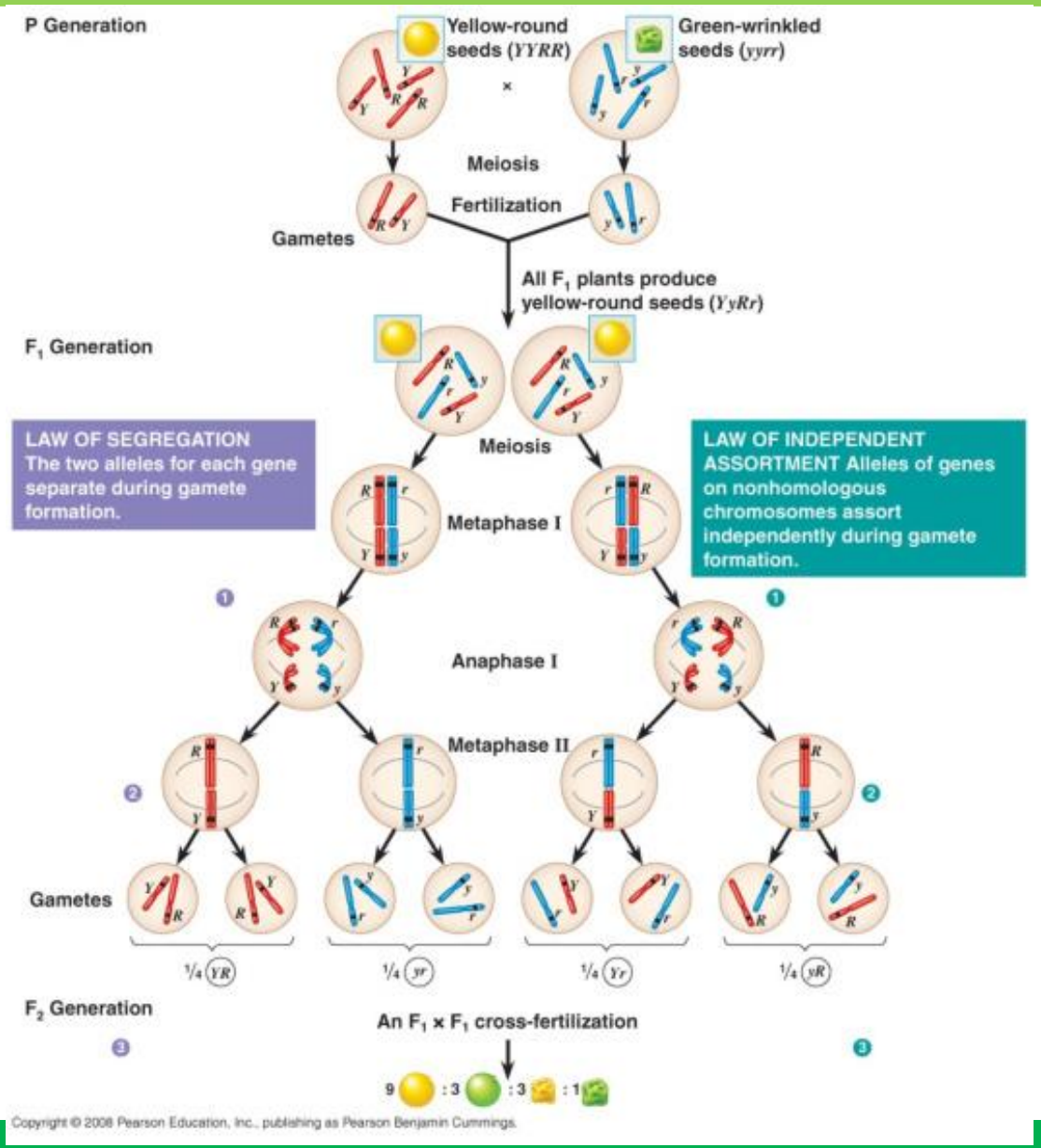
- Mendel's dihybrid cross experiments produced offspring that had a combination of traits that did not match either parent in the P generation.
- If the P generation consists of a yellow-round seed parent (YYRR) crossed with a green-wrinkled seed parent (yyrr), all the F1 plants have yellow-round seeds (YyRr).
- A cross between an F1 plant and a homozygous recessive plant (a testcross) produces four phenotypes.
- Half are the parental types, with phenotypes that match the original P parents, with either yellow-round seeds or green-wrinkled seeds.
- Half are recombinant types or recombinants, new combinations of parental traits, with yellow-wrinkled or green-round seeds.



- The physical basis of recombination between unlinked genes is the random orientation of homologous chromosomes at metaphase I of meiosis, which leads to the independent assortment of the two unlinked genes

Independent Assortment of alleles results in genetic recombination

- A 50% frequency of recombination is observed for any two genes located on different (nonhomologous) chromosomes.
- The physical basis of recombination between unlinked genes is the random orientation of homologous chromosomes at metaphase I of meiosis, which leads to the independent assortment of alleles.
- The F1 parent (YyRr) produces gametes with four different combinations of alleles: YR, Yr, yR, and yr.
- The orientation of the tetrad containing the seed-color gene has no bearing on the orientation of the tetrad with the seed-shape gene.



Recombination of Linked Genes is a result of Crossing Over

In a cross test experiment, the occurrence of parental types with a frequency greater than 50% indicates that the genes are linked.

- Linked genes (genes located on the same chromosome) tend to move together through meiosis and fertilization.
- Under normal Mendelian genetic rules we would not expect linked genes to recombine into assortments of alleles not found in the parents.

– Morgan discovered that genes can be linked, but the linkage was incomplete, as evident from recombinant phenotypes.

- Morgan proposed that some process must sometimes break the physical connection between genes on the same chromosome.
- That mechanism was the **crossing over** of homologous chromosomes.

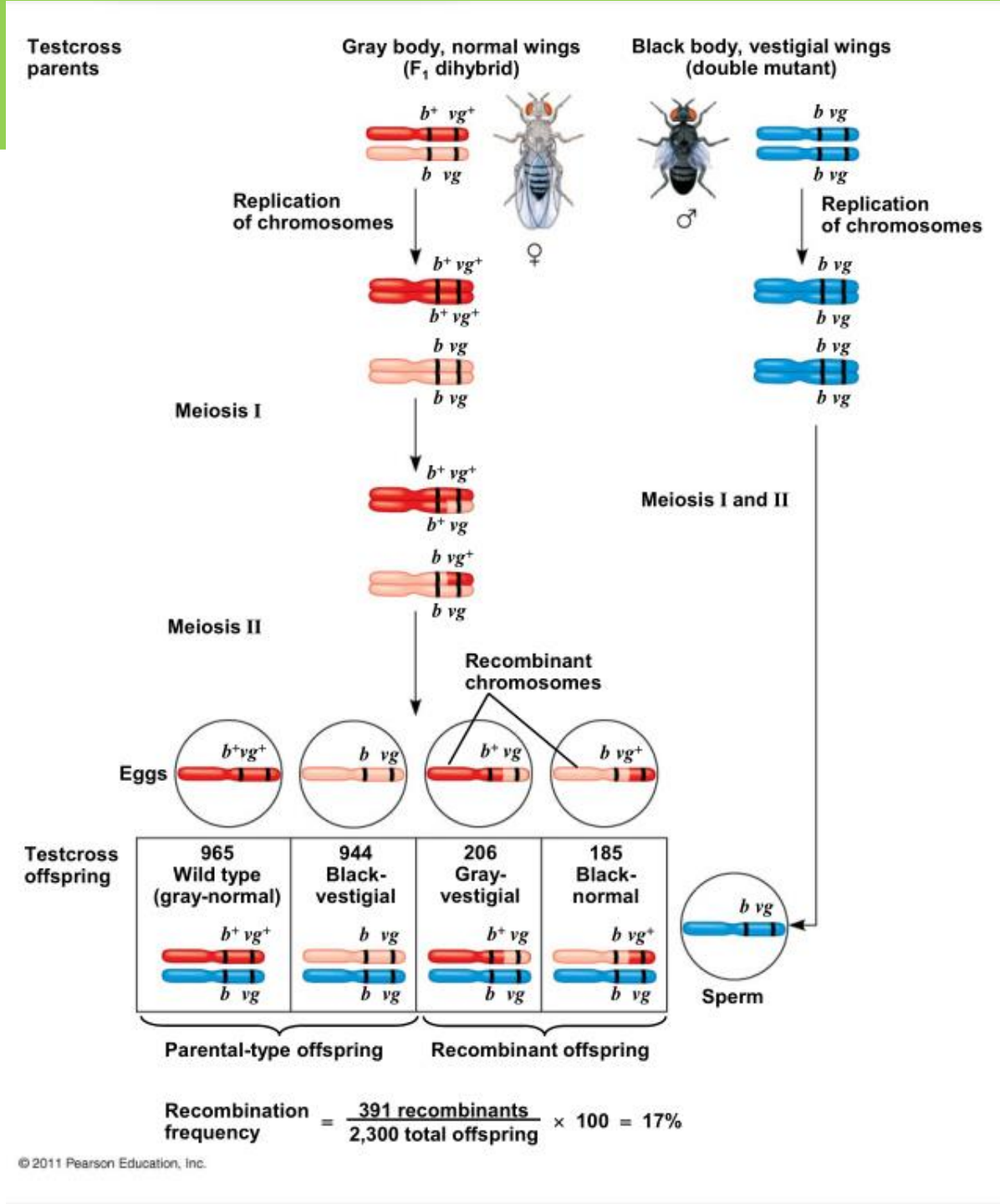
Crossing Over produced recombinant offspring

1. During Meiosis I crossing over between b and vg loci produces new allele combinations in some but not most egg producing cells. Therefore most eggs will harbor the parental type chromosomes.

2. During Meiosis II separation of the chromatids produces recombinant gametes (eggs) with the new allele combinations.

3. Fertilization of the eggs by the bbvgvg sperm will give rise to some recombinant offspring.

The recombination frequency is the percentage of recombinant flies in the total pool of offspring.



New Combinations of Alleles: Variation for Normal Selection

Meiosis and random fertilization generate genetic variation among offspring of sexually reproducing organisms due to :

- independent assortment of chromosomes,
- crossing over in meiosis I, and
- the possibility of any sperm fertilizing any egg.

• Recombinant chromosomes bring alleles together in new combinations in gametes
• Random fertilization increases even further the number of variant combinations that can be produced
• This abundance of genetic variation is the raw material upon which natural selection works. If the traits conferred by particular combinations of alleles are better suited for a given environment, organisms possessing those genotypes will be expected to thrive and leave more offspring, ensuring the continuation of their genetic complement. In the next generation, the alleles will be shuffled anew. Ultimately, the interplay between environment and phenotype (and thus genotype) will determine which genetic combinations persist over time.

Alterations of chromosome number or structure cause some genetic disorders

• Small-scale random mutations are the source of all new alleles and lead to new phenotypic traits.
Large-scale chromosomal changes can also affect an organism's phenotype.

• Physical and chemical disturbances as well as errors during meiosis can damage chromosomes in major ways or alter the number of chromosomes in a cell.

• Large-scale chromosomal alterations often lead to spontaneous abortions (miscarriages) or cause a variety of developmental disorders.

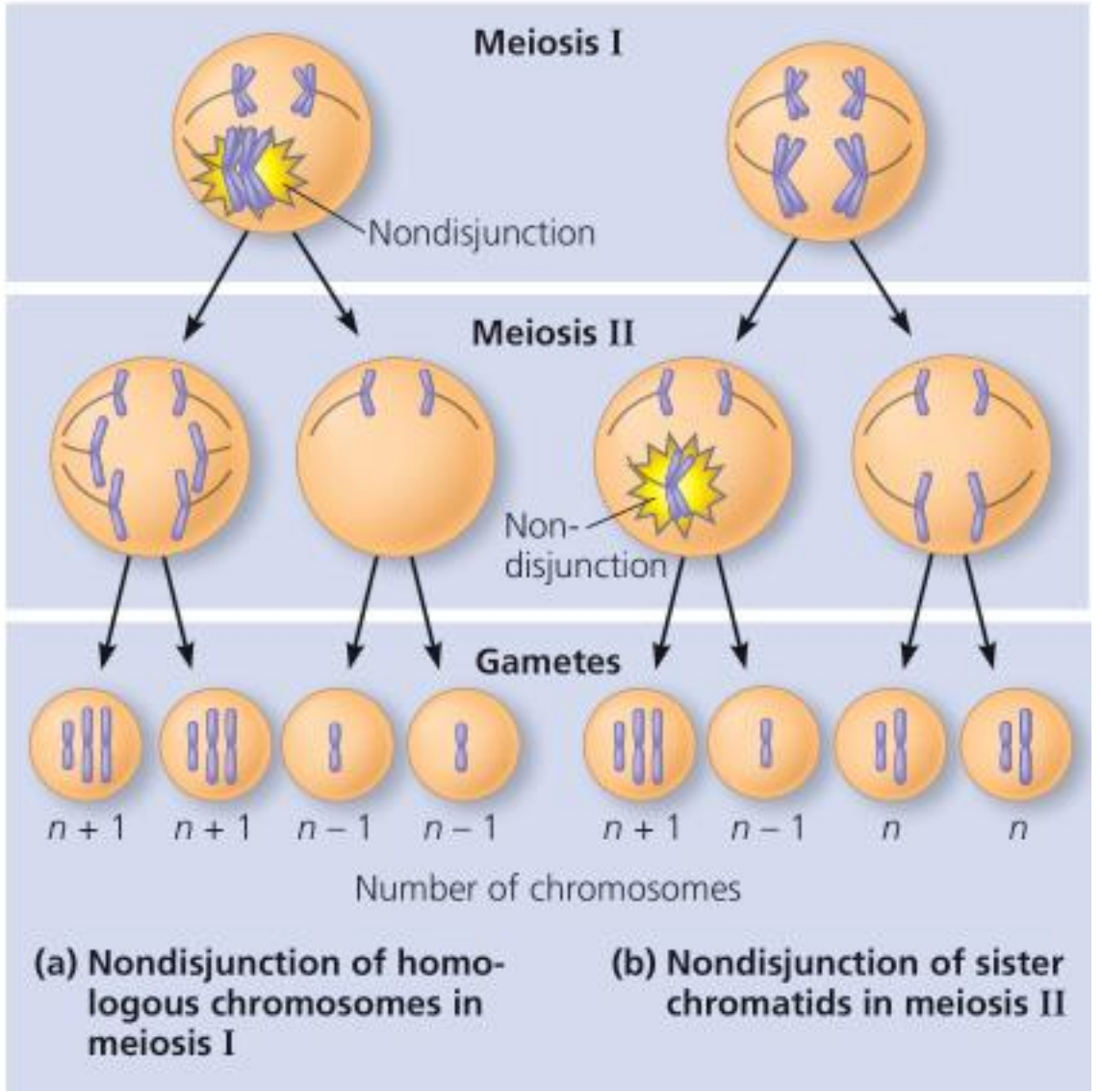
– Plants tolerate genetic defects to a greater extent than do animals.

Abnormal Chromosome Number

Nondisjunction occurs when problems with the meiotic spindle cause errors in separation of homologous chromosomes during meiosis I or separation of sister chromatids during meiosis II.

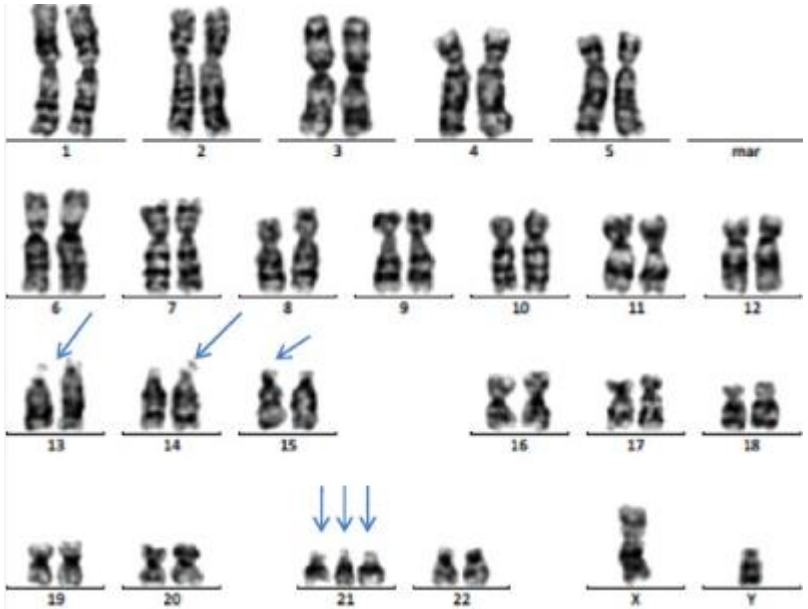
In nondisjunction, one gamete receives two of the same type of chromosome and another
gamete receives no copy. The other chromosomes are usually distributed normally.

▼ **Figure 15.13 Meiotic nondisjunction.** Gametes with an abnormal chromosome number can arise by nondisjunction in either meiosis I or meiosis II. For simplicity, the figure does not show the spores formed by meiosis in plants. Ultimately, spores form gametes that have the defects shown. (See Figure 13.6.)



Offspring produced by nondisjunction

If a gamete with abnormal number of chromosomes unites with a normal gamete in fertilization, the zygote will also have an abnormal number of a particular chromosome, a condition known as **aneuploidy**.



- **Trisomic cells** have three copies of a particular chromosome and have 2n + 1 total chromosomes.
e.g. Down Syndrome (Trisomy 21)

- **Monosomic cells** have only one copy of a particular chromosome and have 2n − 1 chromosomes.
- **Nullisomicsomic cells** have no copies of a particular chromosome and have 2n -2 total chromosomes.
- If the organism survives, aneuploidy typically leads to a distinct phenotype.

Offspring produced by nondisjunction

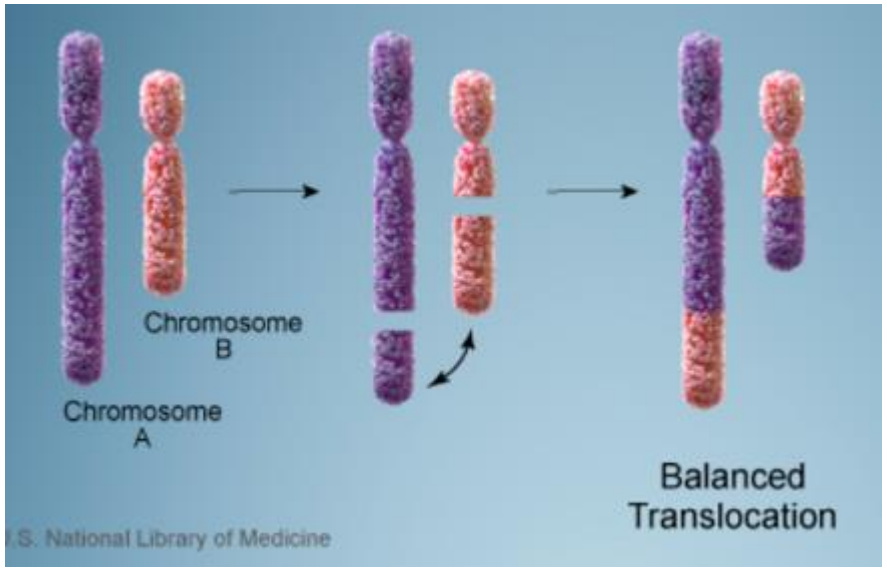
Polyploid organisms have more than two complete sets of chromosomes in all somatic cells.
– **Triploidy**-3 sets (3n) of chromosomes
One way a triploid cell may arise is by the fertilization of an abnormal diploid egg produced by non-disjunction of all its chromosomes.
– **Tetraploidy**-4 sets (4n) of chromosomes. Tetraploidy could result from the failure of a 2n zygote to divide after replicating its chromosomes. Subsequent normal mitotic divisions would
then produce a 4n embryo.

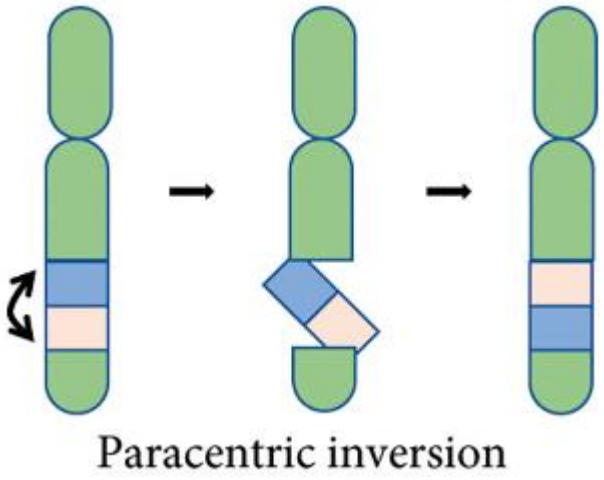
Polyploidy is relatively common among plants and much less common among animals, although it is known to occur in fishes, amphibians and even mammals. e.g. Burrowing rodent
•The spontaneous origin of polyploid individuals plays an important role in the evolution of plants.
•Many crop plants are polyploid. For example, bananas are triploid (3n) and wheat is hexaploid (6n).
•Polyploids are more nearly normal in phenotype than aneuploids.
– One extra or missing chromosome apparently upsets the genetic balance during development more than does an entire extra setof chromosomes.

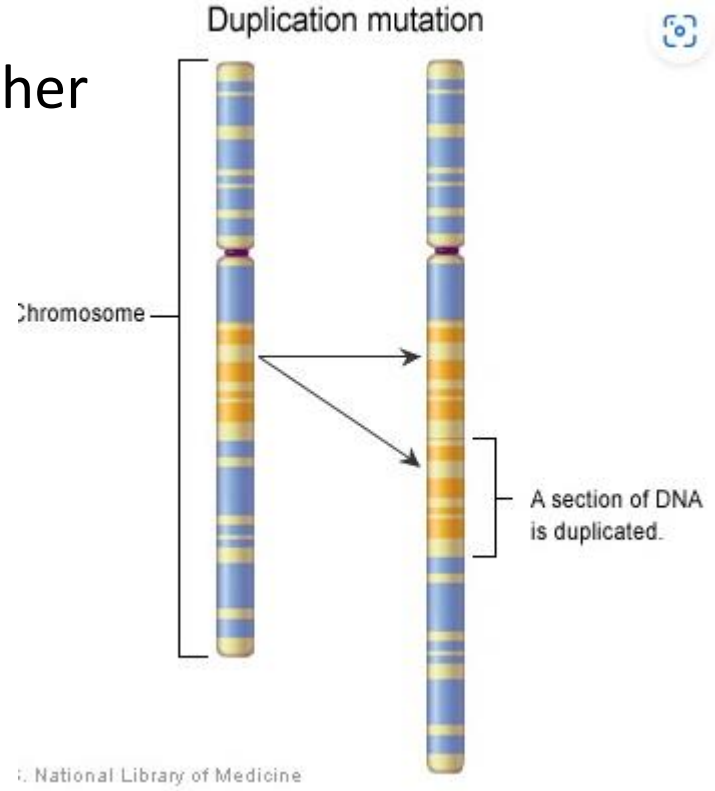
Alterations of Chromosome Structure

Errors in meiosis or damaging agents such as radiation can cause breakage of a chromosome, which can lead to four types of changes in chromosome structure

– Deletion removes a chromosomal segment
– Duplication repeats a segment
– Inversion reverses a segment within a chromosome
– Translocation moves a segment from one chromosome to another

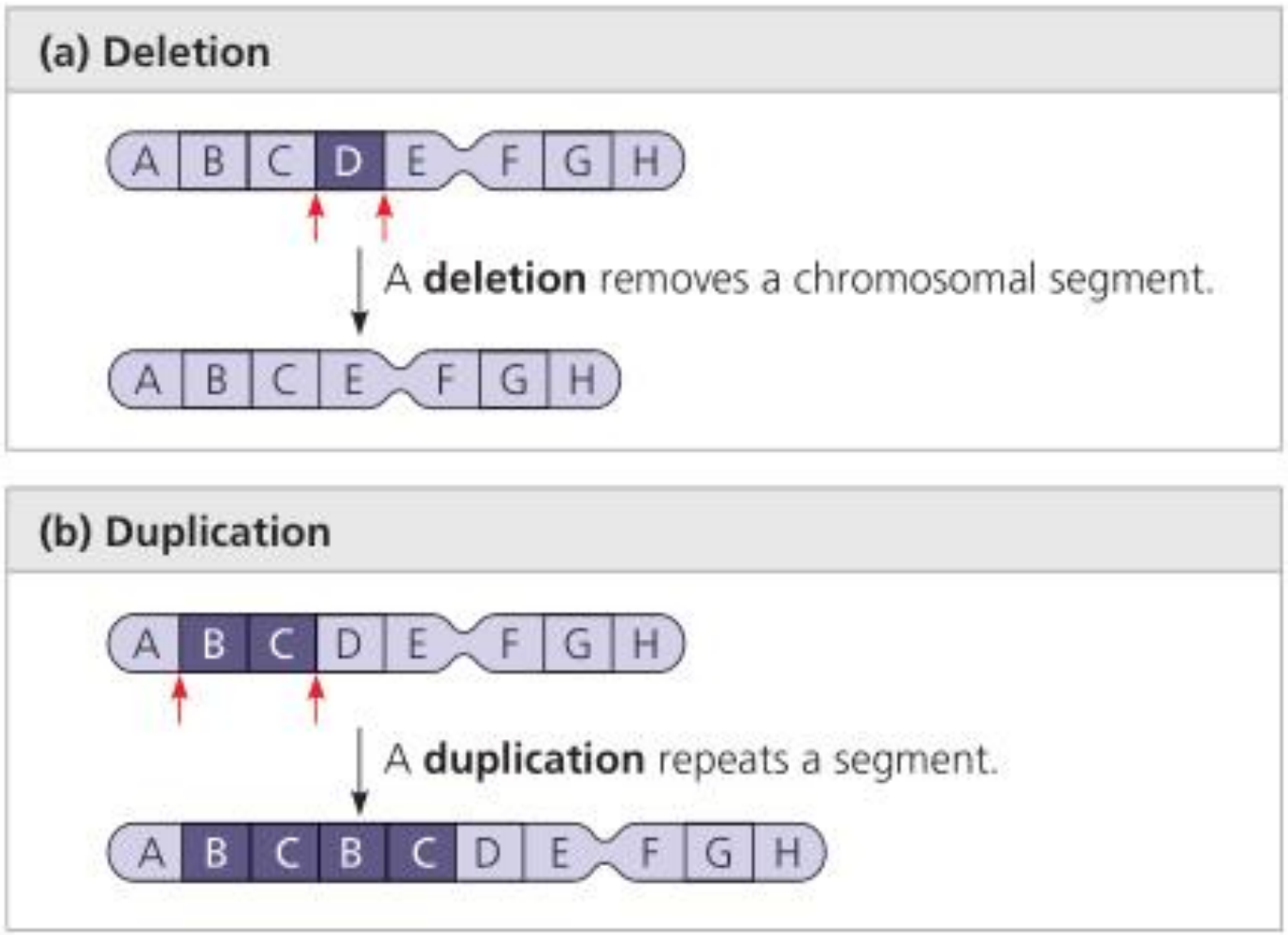






Alterations of Chromosome Structure

▼ Figure 15.14 Alterations of chromosome structure. Red arrows indicate breakage points. Dark purple highlights the chromosomal parts affected by the rearrangements.



Human Disorders Due to Chromosomal Alterations

- Alterations of chromosome number and structure are associated with some serious disorders
  - Most result in miscarriage of the fetus
- Some types of aneuploidy appear to upset the genetic balance less than others, resulting in individuals surviving to birth and beyond
- These surviving individuals have a set of symptoms, or syndrome, characteristic of the type of aneuploidy

Down Syndrome (Trisomy 21)

- Down syndrome is usually the result of an extra chromosome 21, so that each body cell has a total of 47 chromosomes.

- Although chromosome 21 is the smallest human chromosome, trisomy 21 severely alters an individual's phenotype in specific ways.

- The frequency of Down syndrome increases with the age of the mother, a correlation that has not been explained.

Trisomy 21 may be linked to some age-dependent abnormality in a meiosis I checkpoint that normally delays anaphase until all the kinetochores are attached to the spindle.





Aneuploidy of Sex Chromosomes

- Nondisjunction of sex chromosomes produces a variety of aneuploid conditions
- **Klinefelter syndrome** is the result of an extra chromosome in a male, producing **XXY** individuals

–These individuals have male sex organs but abnormally small testes and are sterile.
–Although the extra X is inactivated, some breast enlargement and other female characteristics are common.

- **Monosomy X**, called Turner syndrome, produces **X0** females, who are sterile; it is the only known viable monosomy in humans because their sex organs do not mature.

–When given estrogen replacement therapy, girls with Turner syndrome develop secondary sex characteristics.

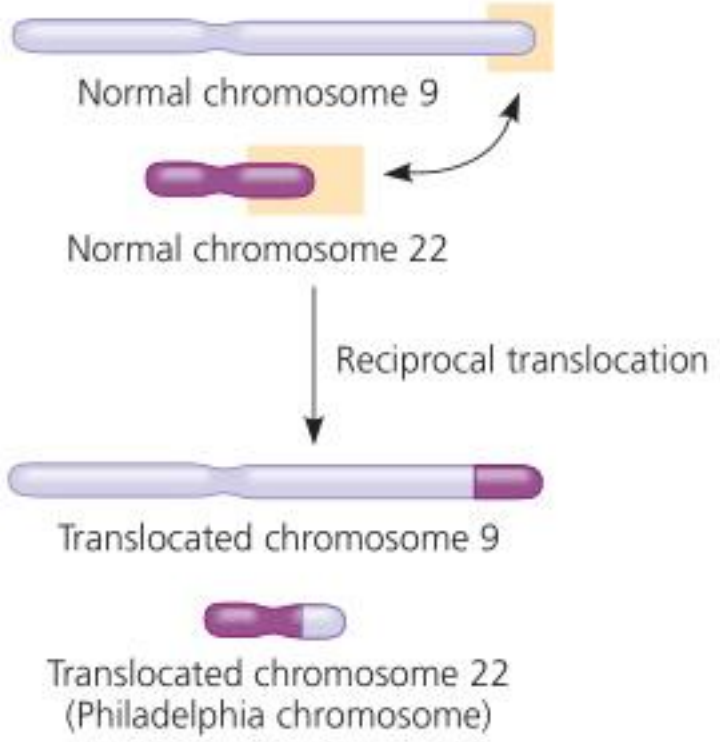
- Trisomy X (XXX) are generally healthy and have no unusual physical features other than being slightly taller than average. Females with trisomy X are at risk for learning disabilities.
- Males with extra Y chromosome (XYY) tend to be somewhat taller than average

Disorders Caused by Structurally Altered Chromosomes

• The syndrome **cri du chat** ("cry of the cat"), results from a specific deletion in chromosome 5
– A child born with this syndrome is mentally retarded and has a catlike cry; individuals usually die in infancy or early childhood.

• Chromosomal translocations can also occur during mitosis; some have been implicated in certain cancers.
i.e. Chronic myelogenous leukemia (CML), are caused by translocations of chromosomes.

▼ **Figure 15.16 Translocation associated with chronic myelogenous leukemia (CML).** The cancerous cells in nearly all CML patients contain an abnormally short chromosome 22, the so-called Philadelphia chromosome, and an abnormally long chromosome 9. These altered chromosomes result from the reciprocal translocation shown here, which presumably occurred in a single white blood cell precursor undergoing mitosis and was then passed along to all descendant cells.



Some inheritance patterns are exceptions to the standard chromosome theory

- There are two normally occuring exceptions to Mendelian genetics:

- One exception involves genes located in the nucleus, and the other exception involves genes located outside the nucleus.

- In both cases, the sex of the parent contributing the allele is a factor in the pattern of inheritance.

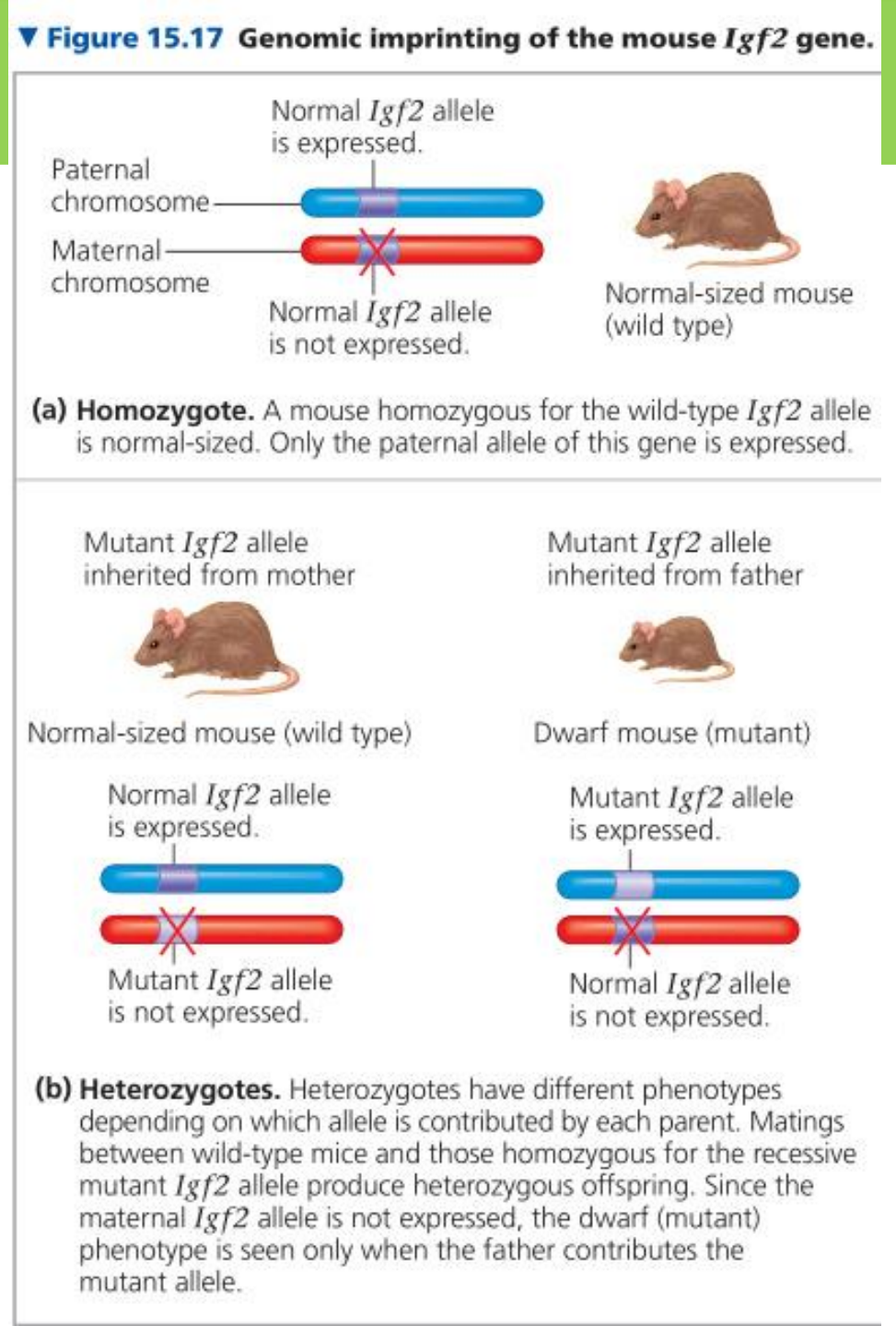
Genomic Imprinting

• Variation in phenotype depending on whether an allele is inherited from the male or female parent is called **genomic imprinting.** (Note that unlike sex-linked genes, most imprinted genes are on autosomes.)
Using newer DNA sequence-based methods, about 100 imprinted genes have been identified in humans.

Genomic imprinting occurs during the formation of gametes and results in the silencing of the imprinted genes.
– Because different genes are imprinted in sperm and eggs, some genes in a zygote are maternally imprinted and others are paternally imprinted.
– For a maternally imprinted gene, only the paternal allele is expressed.
– For a paternally imprinted gene, only the maternal allele is expressed.
•The maternal and paternal imprints are transmitted to all body cells during early development.
•Although only a few genes are imprinted most of these genes are critical for embryonic development.

Genomic Imprinting

- The gene for insulin-like growth factor 2 (Igf2) was one of the first imprinted genes to be identified.
  – Although the growth factor is required for normal prenatal growth, only the paternal allele is expressed.
- Evidence that the Igf2 allele is imprinted initially came from crosses between wild-type mice and dwarf mice homozygous for a recessive mutation in the Igf2 gene.
- The phenotypes of heterozygous offspring differ, depending on whether the mutant allele comes from the mother or the father.
- The Igf2 allele is imprinted in eggs, turning off expression of the imprinted allele.
- In sperm, the Igf2 allele is not imprinted and functions normally.



▼ **Figure 15.17** **Genomic imprinting of the mouse *Igf2* gene.**

(a) Homozygote. A mouse homozygous for the wild-type *Igf2* allele is normal-sized. Only the paternal allele of this gene is expressed.

(b) Heterozygotes. Heterozygotes have different phenotypes depending on which allele is contributed by each parent. Matings between wild-type mice and those homozygous for the recessive mutant *Igf2* allele produce heterozygous offspring. Since the maternal *Igf2* allele is not expressed, the dwarf (mutant) phenotype is seen only when the father contributes the mutant allele.

Genomic Imprinting

In many cases, the imprint seems to consist of methyl (¬CH3) groups that are added to cytosine nucleotides of one of the alleles.

Genomic imprinting is thought to affect only a small fraction of mammalian genes

- Most imprinted genes are critical for embryonic development

Inheritance of Organelle Genes

Not all of a eukaryotic cell's genes are located on nuclear chromosomes, or even in the nucleus; some genes are located in organelles in the cytoplasm. Because they are outside the nucleus, these genes are sometimes called **extranuclear genes** or **cytoplasmic genes**.

Mitochondria, chloroplasts, and other plant plastids carry small circular DNA molecules.

Because a zygote inherits all its mitochondria from egg, all mitochondrial genes in most animals and plants demonstrate maternal inheritance.

– The products of mitochondrial genes make up the protein complexes of the electron transport chain and ATP synthase.
– Some defects in mitochondrial genes prevent cells from making enough ATP and result in diseases that affect the muscular and nervous systems
– For example, mitochondrial myopathy and Leber's hereditary optic neuropathy

Some rare human disorders are produced by mutations to mitochondrial DNA

Tissues that require large energy supplies (the nervous system and muscles) may suffer energy deprivation from these defects.
– For example, a person with mitochondrial myopathy suffers weakness, intolerance of exercise, and muscle deterioration.
– Another mitochondrial disorder is Leber's hereditary optic neuropathy, which can produce sudden blindness in young adults.
• Other mitochondrial mutations may contribute to diabetes, heart disease, and other diseases of aging, such as Alzheimer's disease.
• Over a lifetime, new mutations gradually accumulate in mitochondrial DNA and contribute to the aging process.

- While nuclear DNA is inherited from both parents and recombines with every generation, mitochondrial DNA is inherited solely from your mother and can remain unchanged for tens of thousands of years. As such, mitochondrial DNA ("mitogenome") is a key tool for tracking genetic history.